

# Wirsching's Positive-Predecessor-Density Program: Proofs of Conjectures 1 and 3

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## Abstract

Wirsching (2003) reduces uniform positive predecessor density for the  $3n + 1$  map to a chain of five conditions, organized into three conjectures. We prove two of them. Conjecture 1 concerns his path-counting generators: in the convolution carrying them to his Elka functions the partition weights grow subexponentially while the binomial ratio decays geometrically, so a window of width  $O(\sqrt{\ell})$  dominates and its radius fits inside the hypothesis. Conjecture 3 concerns the asymptotics near 0 of an invariant density  $\varphi$ , a base-3 analogue of the Fabius density, against an explicit  $\varphi_0$  due to Berg and Krüppel. The exact log-Laplace transform of  $\varphi$  splits into a smooth part, a log 3-periodic correction  $H$ , and a doubly exponentially small remainder. Berg and Krüppel represented that correction as an infinite product in 1998; their analysis did not determine whether it is constant. We give  $H$  as a Fourier series with coefficients in closed form in  $\Gamma$  and  $\zeta$ ; a classical zero-free theorem for  $\zeta$  shows it is not constant, and we enclose its oscillation rigorously. Wirsching's comparison class fixes one phase of  $H$ , and there Conjecture 3 holds with limit  $e^{H(0)}$ , numerically  $0.534122\dots$ . Off that class the phase sweeps a full period, so the unrestricted asymptotic  $\varphi(t) \sim \kappa\varphi_0(t)$  fails. Condition  $(\star 4)$  follows, at every window radius, with  $\mu = 1/3$ : what Wirsching's argument needs is weaker than Conjecture 3 itself, and the same saddlepoint chain settles it directly. With Conjecture 1 the chain reduces to the single condition  $(\star 3)$ . Conjecture 2 is his route to it and remains open.

## 1 Introduction

Let  $T$  denote the  $3n + 1$  map on the positive integers,  $T(n) = n/2$  for  $n$  even and  $T(n) = (3n + 1)/2$  for  $n$  odd. Wirsching's monograph [6] develops the predecessor-set machinery this paper descends from, and [1] takes it further, studying the density of predecessor sets through an averaging construction on the 3-adic integers. A family of operators  $S_\ell$ , built from path counts in the predecessor graph, converges in the strong operator topology to a limiting operator whose essential part is a Markov chain on  $[0, 1]$ . That chain has a unique invariant density  $\varphi$ , a base-3 analogue of the Fabius density, and Wirsching reduces uniform positive predecessor density to a chain of five conditions  $(\star 1)$  through  $(\star 5)$  about  $\varphi$  and the path counts, organized into three conjectures. None of the three had a published resolution.

Two of them are proved here.

Conjecture 1 asks that a pointwise lower bound on Wirsching's generators, relative to their 3-adic Haar average, transfer to his Elka functions. Theorem 4.3 proves it. In the convolution carrying the generators to the Elka functions the partition weights grow subexponentially while the binomial ratio decays geometrically, so a window of width  $O(\sqrt{\ell})$  dominates, and its radius

can be budgeted to sit inside the hypothesis. Together with Wirsching’s own two theorems this makes  $(\star 3)$  alone sufficient for uniform positive predecessor density through proved implications only (Corollary 4.4).

Conjecture 3 concerns  $\varphi$  near 0, compared against an explicit  $\varphi_0$ . Wirsching’s equation (7.11) defines  $\varphi_0$  to be Berg and Krüppel’s asymptotic solution [3], naming them in the sentence that introduces it, so the identification is his and not a substitution made here. They had produced  $\varphi_0$  five years earlier, as the asymptotic solution of a truncated version of the functional equation  $\varphi$  satisfies, and gave the periodic correction relating the two as an infinite product without ever evaluating it. The exact log-Laplace transform of  $\varphi$  decomposes as a smooth part, a log 3-periodic correction  $H$ , and a doubly exponentially small remainder (Theorem 6.1). We give  $H$  as a Fourier series with coefficients in closed form in  $\Gamma$  and  $\zeta$  (Proposition 6.3). The classical zero-free theorem for  $\zeta$  on  $\text{Re } s = 1$  then shows  $H$  is not constant, with no computation, and its oscillation is enclosed rigorously (Proposition 6.5). Conjecture 3 is itself restricted: Wirsching asks for the limit “uniformly for sequences  $(z_\ell) \in \tilde{A}_{\delta_5}$ ”, his own comparison class, and that class fixes a single phase of  $H$ . There the limit exists and equals  $e^{H(0)}$ , so **Conjecture 3 is true** (Theorem 10.1). Away from that class the phase sweeps a full period, so the stronger unrestricted asymptotic  $\varphi(t) \sim \kappa\varphi_0(t)$ , which Wirsching considered first and declined, is false (Theorem 10.2). The two are statements about the same ratio at two different scopes, and only the restricted one is Conjecture 3. The function  $e^H$  is Berg and Krüppel’s own periodic factor for this eigenfunction (Corollary 10.3).

Condition  $(\star 4)$  follows, for every window radius, with  $\mu = 1/3$  (Theorem 11.1). Wirsching’s own route there is to compose  $(\star 5) \Rightarrow (\star 4)$  with Conjecture 3, and that works; what his argument actually needs is weaker than Conjecture 3, and Proposition 10.6 settles it directly from the same saddlepoint chain, without the class bookkeeping. His chain is then closed except at  $(\star 3)$ , and Conjecture 2, the implication  $(\star 4) \Rightarrow (\star 3)$ , is the only conjectural link left in it.

Conjecture 2 is not proved here and not attacked here. What can be said about it from the outside is in §12.

§2 places the problem against the density and distributional literature. §3 fixes Wirsching’s objects and states the chain from the primary source. §4 proves Conjecture 1. §§5 to 10 prove Conjecture 3, through the periodic correction, a uniform saddlepoint approximation, an envelope estimate, and the identification with Berg and Krüppel’s asymptotic. §11 composes the chain.

## 2 Related work

The predecessor-density question is older than Wirsching’s formulation of it. Crandall raised the counting of  $3n+1$  predecessors as a quantitative problem [8], and Sander improved his lower bound on the number of descending integers below a given  $x$  [9]. Krasikov then turned the question into a system of difference inequalities on the counting functions [12]. Applegate and Lagarias then produced density exponents by two independent routes: a tree-search method following Crandall [13], and, separately, a linear-programming solution of Krasikov’s system [14]. Krasikov and Lagarias later pushed the same inequalities further, obtaining for each fixed  $a \not\equiv 0 \pmod 3$ , and for  $x$  large depending on  $a$ , at least  $x^{0.84}$  integers below  $x$  with  $a$  in their forward orbit [16]. That is the published record for the exponent. It is a sublinear count with an  $a$ -dependent threshold, so it is weaker in both respects than the uniform positive density, of order  $x/a$ , that Wirsching’s chain targets.

Wirsching’s own contribution to this line begins with an improved predecessor-set estimate [15], continues through the combinatorial structure of those sets [17], the monograph [6] and the urn model [4], and produces in 2003 the five-condition chain studied here [1].

Between the monograph and the chain he tried an analytic route. Wirsching (2001) links an integral operator to the limiting behaviour of transitions between counting functions attached to predecessor sets, reduces the resulting integral equation to a linear retarded functional differential equation, and reports that the normalized averages of the Elka functions diverge pointwise to  $+\infty$ . He states there that what the limiting behaviour of the transition means for the combinatorial quantities themselves was not known [7, §§1,5]. The invariant density  $\varphi$  that the chain's last condition concerns belongs to a function class with its own literature: Berg and Krüppel supplied the asymptotic used here [3], and Rvachev's survey covers the functional-differential family these densities sit in, which Wirsching's 2001 concluding remarks point at and leave open [18, 7]. Operators attached to the Collatz map have also been studied for their own dynamics, where the behaviour is opposite: Béhani, building on earlier work, shows an associated operator on weighted Bergman spaces is hypercyclic and, under conditions, chaotic [19], while the averaging operator  $W_3$  used below contracts each affine hyperplane of fixed mass and fixes exactly one function of unit mass supported in  $[0, 1]$ .

A second line of work models  $3n+1$  dynamics probabilistically, and it is where  $(\star 3)$ 's residue-level content lives. Borovkov and Pfeifer estimated the Syracuse problem through a probabilistic model [20], Kontorovich and Sinai gave a structure theorem for the maps in question [21], and Kontorovich and Lagarias set the stochastic models for  $3x+1$  and  $5x+1$  side by side [22]. Sinai studied the inverse-path ensemble and its random-walk statistics [23], and proved weak convergence of the residues induced modulo  $3^n$  by independent geometric variables to the uniform measure, noting that the strengthened version in which individual probabilities converge is false [24]. Volkov's randomly labelled binary tree for  $5x+1$  carries direction-dependent edge laws [25], and Tao's Syracuse random variable [5] is a closely related canonical residue law, approaching uniformity at fine scales while failing to be uniformly distributed on  $\mathbb{Z}/3^n\mathbb{Z}$ . It is not the object  $(\star 3)$  constrains.  $(\star 3)$  asks for a pointwise lower bound on fixed-cost slices of the generators, at growing 3-adic resolution and uniformly in  $a \in \mathbb{Z}_3^\times$ ; Tao's variable conditions on nothing of the kind. The distance between the two is where the difficulty of  $(\star 3)$  lives.

Current work runs in several directions. Computation has verified the conjecture below  $2^{71}$  [26]. On the exponent,  $x^{0.84}$  [16] remains the published record; a recent preprint returns to the counting problem by a different route and obtains  $x^{0.3227}$ , describing 0.84 as the historical record [10]. A 2026 preprint of Mazur, unrefereed, raises the exponent to  $x^{0.90}$  by an exact feasible point for the level-18 Krasikov–Lagarias linear program together with an adaptive elimination of advanced terms, with the whole theorem formalized in Lean and its two native-computation steps replayed by independent exact verifiers [11]. None of this line reaches what Wirsching's chain targets: these are sublinear counts with an  $a$ -dependent threshold, where the chain aims at a density of order  $x/a$  uniform in  $a$ . Other recent work addresses the finer structure of trajectories, a different question again [27].

The third conjecture sits in a second literature, on the equation  $\varphi$  satisfies rather than on the  $3n+1$  map. That equation is a two-scale difference equation with dilation parameter 3, in Berg and Krüppel's family [3], and  $\varphi$  is a translate of Rvachev's atomic function  $h_3$  [18]. Kato and McLeod's study of  $y'(x) = ay(\lambda x) + by(x)$  [30] is the analytic ancestor. Wirsching noticed the connection to Rvachev's functions himself and left it to future work [7].

Solutions of such equations carry log-periodic fluctuations in their asymptotics. The phenomenon goes back to de Bruijn on Mahler's partition problem [28] and to Erdős and Richmond [29], and the technique of extracting the fluctuation's Fourier coefficients as residues of a Dirichlet series is developed for radix-rational sequences by Dumas [35] and for  $q$ -regular sequences by Heuberger and Krenn [36]. That work treats summatory functions of sequences. Corollary 10.3 applies the same mechanism to an eigenfunction of the difference equation itself, which is where the  $\Gamma$  and  $\zeta$

coefficients of Proposition 6.3 come from, and with them the non-constancy of  $H$ .

We are not aware of a published resolution of any of Wirsching's three conjectures. The 2003 paper is cited sparsely, in bibliographies of the problem and in adjacent work, and we found no published follow-up taking up any of the three. For the second, what this paper gives is an account of how far the machinery natural to it reaches, and where the argument developed here stops.

### 3 Wirsching's chain, from the primary source

Wirsching's central combinatorial objects are the "Elka" functions  $e_\ell(k, a) := |E_{\ell, k}(a)|$ , on domain  $\mathbb{N}_0 \times \mathbb{Z}_3^\times$  [1, §1–2], where  $E_{\ell, k}(a)$  is the set of paths  $b \xrightarrow{T} \dots \xrightarrow{T} a$  in the Collatz graph with  $T = T_0$  exactly  $k$  times and  $T = T_1$  exactly  $\ell$  times ( $T_0, T_1$  the two branches of the accelerated  $3n+1$  map). Their normalized generators  $\tilde{g}_\ell$  (defined below) live on the continuous state space  $\mathbb{X} = \mathbb{I}_3 \times \mathbb{Z}_3^\times$ : the  $\mathbb{Z}_3^\times$  factor carries the 3-adic variable habitat of the Syracuse measure, while the  $\mathbb{I}_3 \cong \{0, 1, 2\}^\mathbb{N}$  factor carries the normalized step count  $k/3^\ell$ , converging to a density  $\varphi$ .

The compactly supported *generators*  $g_\ell(k, a)$  are defined by the recursion [1, §2, eq. (2.1)]

$$g_{\ell+1}(k, a) = \sum_{0 \leq j < 2 \cdot 3^\ell} g_\ell(k - j, \frac{2^{j+1}a-1}{3}), \quad g_0(k, a) = \mathbf{1}[k = 0] \mathbf{1}[a \in \mathbb{Z}_3], \quad (1)$$

linked to the Elka functions by  $e_\ell = p_\ell * g_\ell$ , where  $p_\ell(m)$  counts the ways to pay  $m$  with coins  $c_0 = 1$ ,  $c_j = 2 \cdot 3^{j-1}$  ( $j \geq 1$ ) [1, §2]. The class  $A_\delta$  consists of the integer sequences  $(k_\ell)$  with  $|\ell - k_\ell| \leq \delta\sqrt{\ell}$  for every  $\ell$  [1, eq. (1.5)];  $\tilde{A}_\delta$  is the analogous class of sequences  $(x_\ell) \in \mathbb{I}_3^\mathbb{N}$ , those with  $||3^\ell x_\ell| - \ell| \leq \delta\sqrt{\ell}$  [1, eq. (3.2)], used in conditions  $(\star 4)$  and  $(\star 5)$  below.

Two densities remain.  $\varphi$  satisfies the untruncated equation  $\lambda\varphi'(t) = a(\varphi(at) - \varphi(at - a + 1))$  for  $a = 3$ ,  $\lambda = 2/3$  [1, eq. (7.9)]; the base parameter  $a$  of that equation and of Berg and Krüppel's is unrelated to the 3-adic unit  $a$  everywhere else in this paper.

$\varphi_0$  is the right-hand side of Berg and Krüppel's [3, Proposition 9.1, eq. (9.6)], which they derive as the asymptotic behaviour, as  $t \rightarrow 0^+$ , of a particular solution of the truncated equation  $\lambda g'(t) = ag(at)$ . Their derivation is a saddlepoint calculation whose hypotheses they do not verify: "since the check of these assumptions is only a question of routine, we drop it and restrict ourselves to the necessary formal calculations" [3, §9]. Nothing below depends on that: we take the closed form itself as the *definition* of  $\varphi_0$  and prove every comparison we need for it. At  $a = 3$  and  $\lambda = 2/3$  it reads

$$\ln \varphi_0(t) = \varepsilon \ln(2\beta) - \frac{1}{2} \ln(2\pi) + \gamma \ln t + \delta_{\text{BK}} \ln(-\ln t) - \beta \ln^2\left(\frac{t}{-\ln t}\right), \quad (2)$$

with  $\alpha = \frac{1}{2} - \log_3 2$ ,  $\beta = 1/(2 \ln 3)$ ,  $\delta_{\text{BK}} = \frac{1}{2} + \alpha - 2\beta \ln(2\beta)$ ,  $\gamma = -2\beta - \delta_{\text{BK}} - \frac{1}{2}$  and  $\varepsilon = \frac{1}{2} + \alpha - \beta \ln(2\beta)$ . Proposition 9.1 states the last three;  $\alpha$  and  $\beta$  come from [3, eq. (9.4)], and the specialization  $\lambda = (a-1)/a$  that fixes  $\alpha$  for this equation sits between Propositions 9.2 and 9.3 [3, p. 180], where the printed line carries  $a$  in place of  $\ln a$ ; the value used here is the one (9.4) forces. Their  $\delta$  is written  $\delta_{\text{BK}}$  throughout, since  $\delta$  is the window radius everywhere else here.

Berg and Krüppel expect only that  $\varphi/\varphi_0$  stays bounded and bounded away from zero, weaker than convergence [3, §9], and Wirsching restates the truncated equation and this asymptotic at [1, eq. (7.10)–(7.11)];  $\varphi_0$  is used in Conjecture 3 below in place of  $\varphi$  itself, which has no known closed form.

**Proposition 3.1** (Base-3 Fabius density).  *$\varphi$  is the density of  $X = \sum_{j \geq 1} 2U_j \cdot 3^{-j}$ ,  $U_j$  i.i.d. uniform on  $[0, 1]$  (an identification not in the source), the base-3 analogue of the Fabius density [2] (the base-2 case satisfies  $h'(x) = 4h(2x)$  on  $[0, 1/2]$ ; here  $\varphi'(x) = (9/2)\varphi(3x)$  on  $[0, 2/3]$  by [1, eq. (7.7)], the*

same shape with base 3 in place of base 2: coefficient  $4 = 2^2/(2-1)$  becomes  $9/2 = 3^2/(3-1)$ , support cutoff  $1/2 = (2-1)/2$  becomes  $2/3 = (3-1)/3$ . Extend every function on  $[0, 1]$  by zero to  $\mathbb{R}$  before applying  $W_3 f(x) := \frac{3}{2} \int_{3x-2}^{3x} f(t) dt$ . Among the functions  $f$  with  $\text{supp } f \subset [0, 1]$  and  $\int_0^1 f = 1$ ,  $\varphi$  is the unique fixed point of  $W_3$ , and it is  $C^\infty$ , piecewise polynomial away from the standard Cantor set [1, Corollary 7]. For  $f_0 \in L^1([0, 1])$  with  $\int_0^1 f_0 = 1$ , the iterates  $f_{n+1} := W_3 f_n$  converge geometrically,  $\|f_n - \varphi\|_1 \leq 2^{-n+1} \|f_1 - f_0\|_1$  [1, Corollary 8].

*Proof.* Only the first claim needs an argument; the rest is quoted. Write  $X = \sum_{j \geq 1} 2U_j 3^{-j}$  and  $X' = \sum_{j \geq 1} 2U_{j+1} 3^{-j}$ , so  $X = (2U_1 + X')/3$  with  $X' \stackrel{d}{=} X$  independent of  $U_1$ . Here  $2U_1/3$  is uniform on  $[0, 2/3]$  and independent of  $X'/3$ , so the sum of the two is absolutely continuous. Write  $f$  for its density. Conditioning on  $U_1$  and differentiating,

$$f(x) = \frac{d}{dx} \mathbb{E}_U [F_X(3x - 2U)] = 3 \int_0^1 f(3x - 2u) du = \frac{3}{2} \int_{3x-2}^{3x} f(t) dt = W_3 f(x),$$

the substitution  $t = 3x - 2u$  giving  $du = -dt/2$ . Each summand lies in  $[0, 2 \cdot 3^{-j}]$ , so  $X \in [0, 2 \sum_{j \geq 1} 3^{-j}] = [0, 1]$  and  $\text{supp } f \subset [0, 1]$ ; also  $f \in L^1$  with  $\int_0^1 f = 1$ . So  $f$  is a fixed point of  $W_3$  meeting the two side conditions, and  $\varphi$  is the only one [1, Corollary 7]. Hence  $f = \varphi$ .  $\square$

Wirsching's paper reduces uniform positive predecessor density (that the sets of predecessors of  $a$  have lower asymptotic density bounded below by  $c/a$ , for a single constant  $c > 0$  uniform over admissible targets  $a$ ) to a chain of five conditions [1, Definition 1, §§1–7]:

|                                        |                                                 |
|----------------------------------------|-------------------------------------------------|
| Theorem 1 (proved in [1])              | (★1) $\Rightarrow$ Positive Density             |
| Conjecture 1                           | (★2) $\Rightarrow$ (★1)                         |
| Theorem 2 (stated in [1], proof below) | (★3) $\Rightarrow$ (★2)                         |
| Conjecture 2                           | (★4) $\Rightarrow$ (★3)                         |
| Conjecture 3                           | (★5); (★5) $\Rightarrow$ (★4) argued in [1, §7] |

That is the chain as the source leaves it. Its status after §4 and §10 is the subject of §11: (★5) and (★4) are established outright, every implication from (★3) downward is a theorem, and Conjecture 2 is the one implication still conjectural. Here (★1) is a  $\liminf$  lower bound on Wirsching's Elka functions relative to their Haar average, uniform over sequences  $(k_\ell)$  with  $|\ell - k_\ell| \leq \delta\sqrt{\ell}$ ; (★2) is the same statement for his generators  $g_\ell(k, a)$ ; (★3) upgrades the  $\liminf$  over  $\ell$  to a bound holding for every  $\ell \geq \ell_0$ . Its index is printed rather than implied: Wirsching's Theorem 2 reads “if there are real numbers  $\delta, \mu > 0$  and an index  $\ell_0$  such that...” [1, §3], where (★1) and (★2) leave the eventual index inside a  $\liminf$ . Under the reading adopted above that would make  $\ell_0$  global in  $a$  as well, but  $a$  is free here too and nothing printed forces it; Wirsching's own use of (★3) needs only an  $\ell_0$  per unit. Nothing below turns on which it is; and (★4) asserts that there is a constant  $\mu > 0$  with

$$\liminf_{\ell} \frac{(W_3^\ell \chi_1)(x_\ell)}{(W_3^\ell \chi_0)(x_\ell)} \geq \mu, \quad \text{uniformly for } (x_\ell) \in \tilde{A}_\delta,$$

with  $\chi_0 = \mathbf{1}_{[0, 2/3]}$ ,  $\chi_1 = \mathbf{1}_{[1/3, 1]}$ , and  $W_3$  the averaging operator of Proposition 3.1.

First, (★4) is a statement purely about the one-dimensional operator  $W_3$ . It never mentions Wirsching's generators and never mentions the 3-adic unit  $a$ , so it supplies no residue-level information, which is what (★3) is a statement about. The source offers two bridges: the averaged-generator identity  $\lim_{\ell} \int_{\mathbb{Z}_3^\times} \tilde{g}_\ell(x, a) da = \varphi(x)$  for fixed  $x \in \mathbb{I}_3$  [1, §6], and Wirsching's Theorem 3 [1, §5], strong convergence  $S_\ell \rightarrow S_\infty$  for every  $f \in C(\mathbb{X})$ , of the transition operators  $S_\ell : C(\mathbb{X}) \rightarrow C(\mathbb{X})$  defined

by  $\tilde{g}_\ell = S_\ell(\tilde{g}_{\ell-1})$  [1, §4], uniform on bounded equicontinuous families. Neither bridge is uniform in growing resolution, so neither closes the gap between (★4) and (★3).

Second, (★3) and (★2) are the same inequality, and the implication between them is immediate. Wirsching states Theorem 2 without proof [1, §3], so we give one. Writing  $\tilde{g}_\ell(x, a) = \gamma_\ell g_\ell(\lfloor 3^\ell x \rfloor, a)$  for a constant  $\gamma_\ell > 0$  not depending on  $a$  [1, §3], the constant cancels from both sides of the ratio each condition bounds, so (★3) reads  $g_\ell(\lfloor 3^\ell x_\ell \rfloor, a) \geq \mu \tilde{g}_\ell(\lfloor 3^\ell x_\ell \rfloor)$  for every  $\ell \geq \ell_0$  and every  $(x_\ell) \in \tilde{A}_\delta$ . Given  $(k_\ell) \in A_\delta$ , put  $x_\ell := k_\ell 3^{-\ell}$  wherever that is legitimate. It is not legitimate at every level:  $A_\delta$  admits  $k_\ell$  outside  $[0, 3^\ell - 1]$  at finitely many early levels, where  $k_\ell 3^{-\ell} \notin \mathbb{I}_3$ . Pick  $L(\delta)$  beyond which the whole window lies in  $[0, 3^\ell - 1]$ , which exists since the window grows like  $\sqrt{\ell}$  and the range like  $3^\ell$ . For  $\ell \geq L$  set  $x_\ell := k_\ell 3^{-\ell}$ , so  $\lfloor 3^\ell x_\ell \rfloor = k_\ell$  and  $(x_\ell) \in \tilde{A}_\delta$  by [1, eq. (3.2)]; for  $\ell < L$  take any admissible value. Then (★3) gives the bound at every  $k_\ell$  in the window for  $\ell \geq \max(L, \ell_0)$ , and a bound holding from some index on has  $\liminf$  at least as large, which is (★2) with  $\delta_1 = \delta$  and  $\mu_1 = \mu$ . The finite prefix is invisible to a  $\liminf$ . What (★3) supplies over (★2) is the stronger quantifier, a bound at every  $\ell \geq \ell_0$  rather than in the limit, and the implication just proved discards it.

Third, Theorem 1's conclusion, uniform positive predecessor density, is about  $a \in \mathbb{N}$ , non-cyclic,  $a \not\equiv 0 \pmod{3}$ ; its hypothesis (★1), like (★2) and (★3), is stated over  $\mathbb{Z}_3^\times$ .

## 4 Wirsching's Conjecture 1

Throughout this section, fix  $a \in \mathbb{Z}_3^\times$  and write  $\bar{g}_\ell(k) := \int_{\mathbb{Z}_3^\times} g_\ell(k, b) db$  and  $\bar{e}_\ell(k) := \int_{\mathbb{Z}_3^\times} e_\ell(k, b) db$  for the 3-adic averages. Put  $c_0 = 1$  and  $c_j = 2 \cdot 3^{j-1}$  for  $j \geq 1$ . Wirsching's convolution formulas

$$e_\ell(k, a) = \sum_{j=0}^k p_\ell(k-j) g_\ell(j, a), \quad \bar{e}_\ell = p_\ell * \bar{g}_\ell \quad (3)$$

hold at each fixed  $a$  and on the averages respectively [1, eq. (2.3), (2.4)], with  $p_\ell(m)$  the number of partitions of  $m$  into the parts  $c_0, \dots, c_\ell$  with repetition. Both  $g_\ell$  and  $p_\ell$  are non-negative.

The quantifier order matters here, and the source does not settle it. Wirsching states each of (★1), (★2) and (★3) as “if there are real numbers  $\delta, \mu > 0$  such that...”, with  $a$  occurring free inside the displayed inequality [1, §§1–3]. He never writes the quantifier on  $a$ , so nothing printed distinguishes  $\exists \delta_1, \mu_1 \forall a$  from  $\forall a \exists \delta_1(a), \mu_1(a)$ . We adopt the first, and the reason is his own Definition 1: uniform positive predecessor density there means one constant  $c > 0$  with  $\liminf_n |\{x \in P_T(a) : x \leq n\}|/n \geq c/a$  for each  $a$ , the constant standing outside the quantifier on  $a$  [1, §1]. A  $\mu_1$  free to depend on  $a$  and tend to 0 could not deliver that single  $c$  through his Theorem 1, so the global reading is the only one under which the chain does what he says it does. Stated in full, and understood as a reading rather than as something the source prints:

$$\exists \delta_1, \mu_1 > 0 \quad \forall a \in \mathbb{Z}_3^\times \quad \forall \varepsilon > 0 \quad \exists \ell_0(a, \varepsilon) \quad \forall \ell \geq \ell_0 \quad \forall (k_m) \in A_{\delta_1} : \quad \frac{g_\ell(k_\ell, a)}{\bar{g}_\ell(k_\ell)} \geq \mu_1 - \varepsilon.$$

The stated uniformity is over sequences, so the eventual index  $\ell_0$  may depend on  $a$ ; the constant  $\mu_1$  may not. Below,  $a$  is fixed and only  $\ell_0$  is permitted to depend on it: the argument carries the same  $\mu_1$  from the generators to the Elka functions, for every  $a$ .

**Lemma 4.1** (Window form of the hypothesis). *Suppose (★2) holds with radius  $\delta_1$  and constant  $\mu_1$ . Then for every  $\varepsilon > 0$  there is  $\ell_0$  such that for every  $\ell \geq \ell_0$  and every integer  $j$  with  $|j - \ell| \leq \delta_1 \sqrt{\ell}$ ,*

$$g_\ell(j, a) \geq (\mu_1 - \varepsilon) \bar{g}_\ell(j).$$



*Proof.*  $A_{\delta_1}$  is a product: a sequence lies in it exactly when  $|m - k_m| \leq \delta_1 \sqrt{m}$  at each  $m$  separately [1, eq. (1.5)], with no constraint tying different levels. Write  $W_m$  for the window of admissible indices at level  $m$ , a finite set. On  $0 \leq j \leq 3^m - m - 1$  the average  $\bar{g}_m(j)$  counts urn occupancies summing to  $j$  and is positive, but  $W_m$  need not sit inside that range at small  $m$ : at  $\delta_1 = 1$  and  $m = 1$  the window is  $\{0, 1, 2\}$  while the support stops at 1, and the ratio at  $j = 2$  is  $0/0$ . Since  $W_m$  grows like  $\sqrt{m}$  and the support like  $3^m$ , there is  $L(\delta_1)$  with  $W_m \subset [0, 3^m - m - 1]$  for every  $m \geq L$ . Put

$$k_m^* := \arg \min_{j \in W_m} \frac{g_m(j, a)}{\bar{g}_m(j)} \quad (m \geq L), \quad k_m^* := m \quad (m < L),$$

which is well defined, and  $(k_m^*) \in A_{\delta_1}$ . Applying the hypothesis to this one sequence,  $\liminf_m g_m(k_m^*, a)/\bar{g}_m(k_m^*) \geq \mu_1$ , so there is  $\ell_0$  with  $g_\ell(k_\ell^*, a) \geq (\mu_1 - \varepsilon)\bar{g}_\ell(k_\ell^*)$  for  $\ell \geq \ell_0$ ; enlarge  $\ell_0$  to  $\max(\ell_0, L)$  so that  $k_\ell^*$  is the window argmin there. By the choice of  $k_\ell^*$  every  $j \in W_\ell$  has  $g_\ell(j, a)/\bar{g}_\ell(j) \geq g_\ell(k_\ell^*, a)/\bar{g}_\ell(k_\ell^*)$ , which is the claim.  $\square$

The lemma turns control of one index per level into control of the whole window at one level, which is what the convolution needs. It takes the hypothesis in its weakest reading, one sequence at a time, so nothing below depends on how “uniformly for sequences” is construed.

**Lemma 4.2** (Tail of the convolution). *Fix  $\delta > 0$ . There are absolute  $\rho \in (0, 1)$  and  $A > 0$  such that for every  $\tau > 0$  and all  $\ell$  beyond a threshold depending only on  $\delta$ ,*

$$\frac{1}{\bar{e}_\ell(k)} \sum_{m \geq \tau\sqrt{\ell}} p_\ell(m) \bar{g}_\ell(k - m) \leq \frac{A}{1 - \rho} \rho^{\tau\sqrt{\ell}} \quad \text{uniformly for } |k - \ell| \leq \delta\sqrt{\ell}.$$

*In particular the left side is  $O(e^{-c\sqrt{\ell}})$  with  $c = \tau \log(1/\rho) > 0$ .*

*Proof.* Cancelling  $\prod_{j=0}^\ell (1 - z^{c_j})$  against its inverse in

$$Q_\ell(z) := \sum_{k \geq 0} q_\ell(k) z^k = \prod_{j=0}^\ell \frac{1 - z^{c_j}}{1 - z}, \quad P_\ell(z) := \sum_{m \geq 0} p_\ell(m) z^m = \prod_{j=0}^\ell (1 - z^{c_j})^{-1}$$

gives  $P_\ell(z)Q_\ell(z) = (1 - z)^{-\ell-1}$ , so by (3)

$$\bar{e}_\ell(k) = \frac{1}{2 \cdot 3^{\ell-1}} \binom{k + \ell}{\ell}, \tag{4}$$

where  $q_\ell(k) = 2 \cdot 3^{\ell-1} \bar{g}_\ell(k)$  counts the distributions of  $k$  indistinguishable balls into urns  $U_0, \dots, U_\ell$  with occupancy range  $0, \dots, c_j - 1$  [1, §2], [4, p. 980], which is what the support bound  $\sum_j (c_j - 1) = 3^\ell - \ell - 1$  forces. The urn source writes the maximum occupancy as  $2 \cdot 3^{j-1} - 1 = c_j - 1$ , so  $c_j$  here is that maximum plus one and not the capacity in its sense.

Terms with  $m > k$  contribute nothing:  $g_\ell(j, \cdot) = 0$  for  $j < 0$ , so  $\bar{g}_\ell(k - m) = 0$  there, and the sum may be read over  $0 \leq m \leq k$ . Since  $p_\ell(0) = 1$  and every term is non-negative,  $\bar{g}_\ell(j) \leq \bar{e}_\ell(j)$  for every  $j$ , so by (4)

$$\frac{\bar{g}_\ell(k - m)}{\bar{e}_\ell(k)} \leq \frac{\binom{k - m + \ell}{\ell}}{\binom{k + \ell}{\ell}} = \prod_{i=1}^m \frac{k - i + 1}{k + \ell - i + 1} \leq \left( \frac{k}{k + \ell} \right)^m.$$

In the window  $k \leq \ell + \delta\sqrt{\ell}$ , so  $k/(k + \ell) \leq (\ell + \delta\sqrt{\ell})/(2\ell + \delta\sqrt{\ell}) \rightarrow \frac{1}{2}$  and the ratio is at most  $\rho_0 := 3/5$  for all large  $\ell$ , uniformly in  $k$ . Fix  $\rho := 7/10$ . The partition count obeys for an absolute constant  $C$ , independent of  $\ell$  and  $m$ ,

$$p_\ell(m) \leq \exp(C \log^2(m + 2)) : \quad (5)$$

$p_\ell(0) = 1$ , and for  $m \geq 1$  only the parts  $c_j \leq m$  contribute, at most  $\log_3(m/2) + 2$  of them, each with multiplicity at most  $m$ , so  $p_\ell(m) \leq (m + 1)^{\log_3(m/2) + 2}$ . The bound is uniform in  $\ell$ . Hence  $A := \sup_{m \geq 0} \exp(C \log^2(m + 2))(\rho_0/\rho)^m$  is finite, being the supremum of a subexponentially growing factor against a geometric one with ratio  $6/7 < 1$ , and

$$\sum_{m \geq \tau\sqrt{\ell}} p_\ell(m) \frac{\bar{g}_\ell(k - m)}{\bar{e}_\ell(k)} \leq \sum_{m \geq \tau\sqrt{\ell}} p_\ell(m) \rho_0^m \leq A \sum_{m \geq \tau\sqrt{\ell}} \rho^m = \frac{A}{1 - \rho} \rho^{\lceil \tau\sqrt{\ell} \rceil}. \quad \square$$

**Theorem 4.3** (Wirsching's Conjecture 1). *In the notation of §3, condition  $(\star 2)$ , with window radius  $\delta_1$  (Wirsching's own naming, [1, §2]), implies condition  $(\star 1)$  with some possibly smaller window radius  $\delta < \delta_1$ .*

*Proof.* Choose  $\tau > 0$  and  $\delta > 0$  with  $\delta + \tau < \delta_1$ , which is possible with  $\delta = \tau = \delta_1/3$ , and let  $\varepsilon \in (0, \mu_1)$ . All statements below are for  $\ell$  large enough that  $\tau\sqrt{\ell} \geq 1$  and  $\ell \geq (\delta + \tau)^2$ , so that the retained range is nonempty and  $k - m \geq 0$  throughout it. Take  $\ell_0$  from Lemma 4.1 for that  $\varepsilon$ . Let  $|k - \ell| \leq \delta\sqrt{\ell}$  with  $\ell \geq \ell_0$  large enough for Lemma 4.2. For  $m < \tau\sqrt{\ell}$  the index  $j = k - m$  satisfies  $|j - \ell| \leq (\delta + \tau)\sqrt{\ell} < \delta_1\sqrt{\ell}$ , so Lemma 4.1 applies at every such  $j$  at once. Dropping the non-negative terms with  $m \geq \tau\sqrt{\ell}$  from (3) and then restoring them inside the average,

$$\begin{aligned} e_\ell(k, a) &\geq \sum_{m < \tau\sqrt{\ell}} p_\ell(m) g_\ell(k - m, a) \geq (\mu_1 - \varepsilon) \sum_{m < \tau\sqrt{\ell}} p_\ell(m) \bar{g}_\ell(k - m) \\ &= (\mu_1 - \varepsilon) \left( \bar{e}_\ell(k) - \sum_{m \geq \tau\sqrt{\ell}} p_\ell(m) \bar{g}_\ell(k - m) \right) \geq (\mu_1 - \varepsilon)(1 - \eta_\ell) \bar{e}_\ell(k), \end{aligned}$$

with  $\eta_\ell = O(e^{-c\sqrt{\ell}})$  by Lemma 4.2, uniformly over the window and independent of  $a$ , every quantity entering it being a Haar average. So for every sequence  $(k_\ell) \in A_\delta$ ,  $\liminf_\ell e_\ell(k_\ell, a)/\bar{e}_\ell(k_\ell) \geq \mu_1 - \varepsilon$ , uniformly over such sequences because  $\ell_0$  and  $\eta_\ell$  are. Letting  $\varepsilon \downarrow 0$  gives  $(\star 1)$  at radius  $\delta$  with constant  $\mu_1$ , and  $(\star 1)$  asks only for some positive constant.  $\square$

**Corollary 4.4.** *Condition  $(\star 3)$  implies uniform positive predecessor density on the non-cyclic integers  $a \not\equiv 0 \pmod{3}$ .*

*Proof.* Wirsching's Theorem 2 takes  $(\star 3)$  to  $(\star 2)$ , stated in [1, §3] and proved in §3 above; Theorem 4.3 takes  $(\star 2)$  to  $(\star 1)$ , and his Theorem 1 takes  $(\star 1)$  to the conclusion [1, §1]. Both of his theorems quantify their window radius and constant existentially, so the possibly smaller radius produced by Theorem 4.3 is admissible in Theorem 1 as it stands.  $\square$

Before this the chain from  $(\star 3)$  downward passed through Conjecture 1. It no longer does, and  $(\star 4) \Rightarrow (\star 3)$  is the only conjectural link left below  $(\star 4)$ .

*Remark 4.5* (Relation to the source). Wirsching records the partition-counting identity  $\prod_{j=0}^\ell (1 - z^{c_j})^{-1} = \sum_m p_\ell(m) z^m$  [1, §2, eq. (2.2)] and the closed form for  $\bar{e}_\ell(k)$  used above [1, §1, eq. (1.4)], and describes, without a generating function, the urn count  $q_\ell(k) = 2 \cdot 3^{\ell-1} \bar{g}_\ell(k)$ , crediting the combinatorial identification to [4] [1, §2]; the generator apparatus itself is developed in [6, Ch. IV,



p. 103 and Thm. IV.1.14], the pinpoints Wirsching himself gives for it [1, §2]; the generating function  $\sum_k q_\ell(k)z^k = \prod_{j=0}^\ell (1 - z^{c_j})/(1 - z)$  used above does not appear there. Wirsching states the implication itself as Conjecture 1 [1, §§1–2]. The cancellation  $P_\ell(z)Q_\ell(z) = (1 - z)^{-\ell-1}$  is what makes  $\bar{e}_\ell(k)$  binomial.

The second conjecture is untouched by this. Its antecedent  $(\star 4)$  comes from Conjecture 3, an asymptotic assertion on  $\varphi(z_\ell)/\varphi_0(z_\ell)$  along the central-limit window  $|\ell - k_\ell| \leq \delta_5 \sqrt{\ell}$ , which §§5 to 10 prove and §11 composes with the present section.

## 5 The invariant density and its Laplace transform

Proposition 3.1 identifies  $\varphi$  as the density of  $X = \sum_{j \geq 1} 2 \cdot 3^{-j} U_j$  with  $U_j$  i.i.d. uniform on  $[0, 1]$ , the unique fixed point of  $W_3$  of unit mass supported in  $[0, 1]$ . At dilation parameter  $a = 3$  and  $\lambda = 2/3$  it is an instance of Berg and Krüppel’s family  $\lambda \psi'(t) = a(\psi(at) - \psi(at - a + 1))$ , and a translate of Rvachev’s atomic function  $h_3$  [18]. Three further facts are used throughout below.

The reflection  $x \mapsto 1 - x$  relating  $\varphi$  to [1, Cor. 7]’s fixed point is vacuous:  $\varphi(x) = \varphi(1 - x)$  identically, since  $X$  and  $1 - X = \sum_{j \geq 1} \frac{2}{3^j} (1 - U_j)$  are equidistributed, because  $1 - U_j \sim U_j$  independently for every  $j$  and  $\sum_{j \geq 1} 2 \cdot 3^{-j} = 1$ . It is continuous, by Lemma 7.3 below. And it is constant, equal to  $3/2$ , on the middle third  $[1/3, 2/3]$ : there  $3x - 2 \leq 0 \leq 1 \leq 3x$ , so  $\text{supp } \varphi \subset [0, 1]$  collapses the integral in  $W_3 \varphi = \varphi$  to  $\int_0^1 \varphi = 1$ .

For  $s > 0$  write

$$K(s) = \log \mathbb{E}[e^{-sX}] = \sum_{j \geq 1} g\left(\frac{2s}{3^j}\right), \quad g(b) = \log \frac{1 - e^{-b}}{b}.$$

The branch of  $g$  is the one analytic on  $\{\text{Re } w > 0\}$  and real on  $(0, \infty)$ ; since  $(1 - e^{-w})/w$  is analytic and non-vanishing there (its zeros lie on  $2\pi i\mathbb{Z} \setminus \{0\}$ , and  $w = 0$  is a removable singularity with value 1), and the half-plane is simply connected,  $g$  exists and is unique. Differentiating termwise, justified because  $g(w) = -w/2 + O(w^2)$  as  $w \rightarrow 0$  so the tail of the series is normally convergent together with every derivative,

$$t(s) := -K'(s), \quad V(s) := s^2 K''(s) > 0,$$

the second an exact positivity,  $K''(s)$  being the variance of  $X$  under its exponential tilt by  $e^{-sX}$ . Since  $t'(s) = -K''(s) < 0$ ,  $t$  is strictly decreasing on  $(0, \infty)$ ; at the endpoints,  $t(0^+) = \mathbb{E}[X] = \sum_{j \geq 1} 2 \cdot 3^{-j} \cdot \frac{1}{2} = \frac{1}{2}$ , and  $t(s) \rightarrow 0$  as  $s \rightarrow \infty$  because the exponential tilt concentrates on  $\text{ess inf } X = 0$  (each factor  $e^{-sX}$  suppresses any mass away from 0 as  $s \rightarrow \infty$ , and  $X \geq 0$  a.s.). So  $t(s)$  decreases strictly from  $1/2$  to 0, and studying  $\varphi(t(s))$  as  $s \rightarrow \infty$  is the same as studying  $\varphi(t)$  as  $t \rightarrow 0^+$ .

The comparison function  $\varphi_0$  is (2), Berg and Krüppel’s asymptotic solution of the truncated equation  $\lambda \psi'(t) = a \psi(at)$  by Laplace transform and saddlepoint analysis, with the constants  $\alpha, \beta, \gamma, \delta_{\text{BK}}, \varepsilon$  fixed there. Their dilation parameter  $a$  is reused in §6 for this paper’s own, unrelated linear coefficient of  $Q$ ; the two never appear in the same formula except in §9, where the collision is flagged at the point it matters. Write  $\varphi_{0, \text{bare}}$  for (2) without the leading constant  $(2\beta)^\varepsilon / \sqrt{2\pi}$ .

## 6 The periodic correction

Set  $c = \log 3$  and write  $w = \log s$ ,  $L(w) = K(e^w)$ . Substituting into the defining series and splitting off the  $j = 1$  term against the rest reproduces the recurrence

$$L(w + c) - L(w) = \log(1 - e^{-2e^w}) - w - \log 2. \quad (6)$$

Let

$$Q(w) = -\frac{w^2}{2c} + \left(\frac{1}{2} - \frac{\log 2}{c}\right)w,$$

chosen so that  $Q(w+c) - Q(w) = -w - \log 2$ , matching (6)'s non-periodic terms exactly. Write  $a := \frac{1}{2} - \frac{\log 2}{c}$  for  $Q$ 's linear coefficient, used again from Section 8 onward.

**Theorem 6.1.** *Define*

$$H(w) := L(w) - Q(w) + \sum_{k \geq 0} \log(1 - e^{-2 \cdot 3^k e^w}). \quad (7)$$

*Then  $H$  is exactly  $c$ -periodic,  $H(w+c) = H(w)$  for every  $w \in \mathbb{R}$ , and*

$$L(w) = Q(w) + H(w) + \Delta(w), \quad \Delta(w) = -\sum_{k \geq 0} \log(1 - e^{-2 \cdot 3^k e^w}),$$

*with  $\Delta^{(j)}(w) = O_j(e^{jw-2e^w})$  for every  $j \geq 0$ : doubly exponentially small, together with all its derivatives, as  $w \rightarrow +\infty$ .*

*Proof.* Write  $d(w) = \log(1 - e^{-2e^w})$ , so (6) reads  $L(w+c) - L(w) = d(w) - w - \log 2$ . Directly from the definition of  $Q$  (expand  $(w+c)^2 = w^2 + 2wc + c^2$  and collect terms),  $Q(w+c) - Q(w) = -w - \log 2$  as well, so the two non-periodic parts cancel exactly:

$$(L - Q)(w+c) - (L - Q)(w) = [d(w) - w - \log 2] - [-w - \log 2] = d(w).$$

The series in (7) telescopes this exactly: with  $D(w) := (L - Q)(w)$ ,

$$D(w) + \sum_{k \geq 0} d(w + kc) = D(w+c) - d(w) + \sum_{k \geq 0} d(w + kc) = D(w+c) + \sum_{k \geq 0} d(w + (k+1)c),$$

so the right side of (7) is invariant under  $w \mapsto w+c$ , that is,  $H$  is  $c$ -periodic. Convergence of the series and its termwise differentiability, together with the stated bound on  $\Delta^{(j)}$ , follow from  $d(w) = O(e^{-2e^w})$  and the corresponding bounds on its derivatives, since  $2 \cdot 3^k e^w$  grows like  $3^k$  for  $k \geq 0$ : the terms  $d(w+kc)$  decay doubly exponentially in  $k$ , hence faster than any fixed geometric rate, which gives uniform, dominated convergence of the series and its derivatives for  $w$  in any half-line bounded away from  $-\infty$ .  $\square$

Given  $Q$ ,  $H$  is uniquely determined by the requirement that  $L - Q - H$  be  $o(1)$  as  $w \rightarrow \infty$ : any two periodic functions differing from  $L - Q$  by an  $o(1)$  term as  $w \rightarrow \infty$  must be equal, being periodic functions that agree in the limit along every residue class of  $w$  modulo  $c$ .

*Remark 6.2.* Berg and Krüppel's own Proposition 9.3, specialized to their dilation parameter 3 and their  $b = 3/2$  (exactly this eigenfunction; their  $\alpha$ , kept as a symbol here, is exactly Section 5's imported  $\alpha$ , which Proposition 9.1 below identifies with this paper's own  $a$ ), gives the equivalent identity  $\exp(H(w)) = 3^{\frac{1}{2}t^2 - \alpha t} \prod_{k \geq 0} \frac{1 - e^{-3^{t-k}/b}}{3^{t-k}/b} \prod_{i \geq 1} (1 - e^{-3^{t+i}/b})$ ,  $t = w/c$ , as an infinite product, in their proof of that proposition. This is  $e^{-Q} e^L e^{-\Delta}$  term for term: writing  $s = e^w = 3^t$ ,  $3^{t-k}/b = 2s/3^{k+1}$ , so reindexing  $j := k+1 \geq 1$  turns the first product into  $\prod_{j \geq 1} \frac{1 - e^{-2s/3^j}}{2s/3^j}$ , exactly  $e^{L(w)}$  from  $K$ 's defining series;  $3^{t+i}/b = 2s \cdot 3^{i-1}$ , so reindexing  $k := i-1 \geq 0$  turns the second into  $\prod_{k \geq 0} (1 - e^{-2s \cdot 3^k})$ , exactly  $e^{-\Delta(w)}$  by (7); and, since  $w = ct$  and  $e^c = 3$ ,

$$-Q(w) = \frac{w^2}{2c} - aw = c(\frac{1}{2}t^2 - at), \quad \text{so} \quad 3^{\frac{1}{2}t^2 - \alpha t} = e^{-Q(w)}$$

once  $\alpha = a$  (Proposition 9.1 identifies Berg and Krüppel's  $\alpha$  with this paper's own linear coefficient of  $Q$ , also called  $a$ ). So Proposition 6.5's primary computation from (12) below is, in substance, an evaluation of this product. What Proposition 6.3's Fourier series supplies instead, and their product form does not, are the derivative bounds (10), on which the rest of the paper's saddlepoint estimates depend.

**Proposition 6.3.**  *$H$  has the Fourier expansion  $H(w) = \sum_{m \in \mathbb{Z}} \hat{H}(m) e^{i\omega_m w}$ ,  $\omega_m = 2\pi m/c$ , with*

$$\hat{H}(0) = \frac{\log 2}{2} - \frac{\log^2 2}{2c} - \frac{c}{12} - \frac{A}{c}, \quad A = \frac{\pi^2}{12} - \frac{\gamma_E^2}{2} - \gamma_1,$$

$$\hat{H}(m) = -\frac{2^{i\omega_m}}{c} \Gamma(-i\omega_m) \zeta(1 - i\omega_m), \quad m \neq 0,$$

where  $\gamma_E$  is the Euler–Mascheroni constant and  $\gamma_1$  the first Stieltjes constant in the convention  $\zeta(s) = \frac{1}{s-1} + \sum_{n \geq 0} \frac{(-1)^n}{n!} \gamma_n (s-1)^n$ , so  $\gamma_1 = -0.0728158\dots$ . The sign convention matters: the other one moves  $\hat{H}(0)$  by about 0.133.

*Proof.* Both  $g$  and  $K$  have Mellin transforms on the strip  $-1 < \operatorname{Re} z < 0$ . As  $b \rightarrow 0^+$ ,  $g(b) = -b/2 + O(b^2)$ ; as  $b \rightarrow \infty$ ,  $g(b) = -\log b + O(e^{-b})$ . So  $\int_0^\infty |g(b)| b^{\sigma-1} db < \infty$  for  $-1 < \sigma < 0$ , and the same two bounds applied termwise give  $K(s) = O(s)$  as  $s \rightarrow 0^+$  and  $K(s) = O(\log^2 s)$  as  $s \rightarrow \infty$ : the terms with  $3^j \leq 2s$  number  $O(\log s)$  and are each  $O(\log s)$ , and the terms with  $3^j > 2s$  sum to  $O(1)$ .

Integrate by parts. With  $g'(b) = 1/(e^b - 1) - 1/b$ , and both boundary terms vanishing on the strip ( $g(b)b^z \rightarrow 0$  at 0 because  $\operatorname{Re} z > -1$ , and at  $\infty$  because  $\operatorname{Re} z < 0$ ),

$$g^*(z) = \int_0^\infty g(b) b^{z-1} db = -\frac{1}{z} \int_0^\infty \left( \frac{1}{e^b - 1} - \frac{1}{b} \right) b^z db = -\frac{\Gamma(z+1)\zeta(z+1)}{z} = -\Gamma(z) \zeta(1+z),$$

the middle step being the classical continuation  $\int_0^\infty \left( \frac{1}{e^u - 1} - \frac{1}{u} \right) u^{v-1} du = \Gamma(v)\zeta(v)$ , valid for  $0 < \operatorname{Re} v < 1$ , at  $v = z+1$ .

Now the harmonic sum. In the  $j$ -th term substitute  $u = 2s/3^j$ , so  $s = 3^j u/2$  and  $s^{z-1} ds = (3^j/2)^z u^{z-1} du$ :

$$\int_0^\infty g\left(\frac{2s}{3^j}\right) s^{z-1} ds = \left(\frac{3^j}{2}\right)^z \int_0^\infty g(u) u^{z-1} du = \left(\frac{3^j}{2}\right)^z g^*(z).$$

Summing over  $j \geq 1$  is legitimate termwise, since  $\sum_{j \geq 1} (3^j/2)^\sigma \int_0^\infty |g(u)| u^{\sigma-1} du$  converges for  $-1 < \sigma < 0$ . The multiplier that appears is therefore

$$\sum_{j \geq 1} \left(\frac{3^j}{2}\right)^z = 2^{-z} \sum_{j \geq 1} 3^{jz} = \frac{2^{-z} 3^z}{1 - 3^z} = \frac{(3/2)^z}{1 - 3^z},$$

a geometric series of ratio  $3^z$ , of modulus  $3^{\operatorname{Re} z} < 1$ , convergent on  $\operatorname{Re} z < 0$  and nowhere else. Hence

$$K^*(z) = \frac{(3/2)^z}{1 - 3^z} \cdot (-\Gamma(z)\zeta(1+z)) = \frac{(3/2)^z \Gamma(z) \zeta(1+z)}{3^z - 1}, \quad -1 < \operatorname{Re} z < 0. \quad (8)$$

The dilations  $\mu_j = 2/3^j$  enter inverted, through  $\mu_j^{-z}$ , which is where the  $2^{-z}$  comes from, and with it the  $2^{i\omega_m}$  of the statement.

The restriction to  $\operatorname{Re} z > -1$  is not cosmetic:  $z = -1$  is a genuine pole of  $K^*$ , so the shift cannot be taken further left. In  $\operatorname{Re} z > -1$  the right side of (8) is meromorphic, with a pole of order three at  $z = 0$  (each of  $\Gamma(z)$ ,  $\zeta(1+z)$  and  $(3^z - 1)^{-1}$  contributing one) and a simple pole at each remaining zero  $z = i\omega_m$ ,  $m \neq 0$ , of  $3^z - 1$ . To the right of the imaginary axis it is analytic.

Fix  $\sigma \in (-1, 0)$  and  $\sigma' \in (0, 1)$ , and shift the inversion contour in  $K(s) = (2\pi i)^{-1} \int_{(\sigma)} K^*(z) s^{-z} dz$  from  $\sigma$  to  $\sigma'$ , using rectangles with horizontal sides at  $\operatorname{Im} z = \pm T_M$ ,  $T_M := \pi(2M+1)/c$ . On those sides  $3^z = 3^{\operatorname{Re} z} e^{i\pi(2M+1)} = -3^{\operatorname{Re} z}$ , so  $|3^z - 1| = 1 + 3^{\operatorname{Re} z} \geq 1$ ; Stirling makes  $|\Gamma(z)|$  decay like  $e^{-\pi T_M/2}$  times a power of  $T_M$ , uniformly for  $\sigma \leq \operatorname{Re} z \leq \sigma'$ , and  $\zeta(1+z)$  grows at most polynomially there. The horizontal contributions therefore vanish as  $M \rightarrow \infty$ , the sum of residues converges absolutely, and on the right vertical line  $|3^z - 1| \geq 3^{\sigma'} - 1 > 0$  makes  $\int_{(\sigma')} |K^*(z) s^{-z}| |dz| = O(s^{-\sigma'})$ . So

$$K(s) = -\operatorname{Res}_{z=0} [K^*(z) s^{-z}] - \sum_{m \neq 0} \operatorname{Res}_{z=i\omega_m} [K^*(z) s^{-z}] + O(s^{-\sigma'}).$$

Take the simple poles first. At  $z = i\omega_m$  one has  $3^{i\omega_m} = e^{2\pi i m} = 1$ , so  $(3^z - 1)' = 3^z \log 3$  equals  $c$  there and  $(3/2)^{i\omega_m} = 3^{i\omega_m} 2^{-i\omega_m} = 2^{-i\omega_m}$ . With  $s = e^w$ ,

$$-\operatorname{Res}_{z=i\omega_m} [K^*(z) e^{-zw}] = -\frac{2^{-i\omega_m}}{c} \Gamma(i\omega_m) \zeta(1 + i\omega_m) e^{-i\omega_m w},$$

and replacing  $m$  by  $-m$  throughout the sum turns the whole family into  $\hat{H}(m) e^{i\omega_m w}$  with  $\hat{H}(m)$  exactly as stated.

At the origin, insert

$$\Gamma(z) = \frac{1}{z} - \gamma_E + \left( \frac{\gamma_E^2}{2} + \frac{\pi^2}{12} \right) z + O(z^2), \quad \zeta(1+z) = \frac{1}{z} + \gamma_E - \gamma_1 z + O(z^2).$$

The two  $1/z$  terms cancel against each other and

$$\Gamma(z) \zeta(1+z) = \frac{1}{z^2} + A + O(z), \quad A = \frac{\pi^2}{12} - \frac{\gamma_E^2}{2} - \gamma_1,$$

which is where  $A$  comes from. Also  $(3^z - 1)^{-1} = (cz)^{-1} - \frac{1}{2} + cz/12 + O(z^3)$ , and  $(3/2)^z e^{-zw} = e^{\theta z}$  with  $\theta := \log(3/2) - w$ . Multiplying the three expansions and reading off the coefficient of  $z^{-1}$ ,

$$\operatorname{Res}_{z=0} [K^*(z) e^{-zw}] = \frac{1}{c} \cdot \frac{\theta^2}{2} - \frac{\theta}{2} + \frac{c}{12} + \frac{A}{c}.$$

Substitute  $\theta = \log(3/2) - w$  and  $\log(3/2) = c - \log 2$ . The  $w$ -dependent part of  $-\operatorname{Res}$  is

$$-\frac{w^2}{2c} + \left( \frac{\log(3/2)}{c} - \frac{1}{2} \right) w = -\frac{w^2}{2c} + \left( \frac{1}{2} - \frac{\log 2}{c} \right) w = Q(w),$$

and the constant part is  $\frac{1}{2} \log 2 - \frac{\log^2 2}{2c} - \frac{c}{12} - \frac{A}{c} = \hat{H}(0)$ .

Collecting, with  $s = e^w$ ,

$$L(w) = Q(w) + \tilde{H}(w) + O(e^{-\sigma' w}), \quad \tilde{H}(w) := \hat{H}(0) + \sum_{m \neq 0} \hat{H}(m) e^{i\omega_m w}.$$

That series converges absolutely and uniformly, because  $|\Gamma(-i\omega_m)| = \sqrt{\pi/(\omega_m \sinh \pi \omega_m)}$  decays like  $e^{-\pi \omega_m/2}$  while  $\zeta(1 - i\omega_m)$  grows polynomially, so  $\tilde{H}$  is continuous and exactly  $c$ -periodic. It is real valued:  $\Gamma$  and  $\zeta$  are real on the real axis, so  $\hat{H}(-m)$  is the complex conjugate of  $\hat{H}(m)$ .

The remainder  $O(e^{-\sigma'w})$  is only ordinarily exponentially small, weaker than the doubly exponential  $\Delta$  of Theorem 6.1, and it is enough. Theorem 6.1 gives  $L - Q - H = \Delta = o(1)$  as  $w \rightarrow +\infty$ , so  $\tilde{H} - H = O(e^{-\sigma'w}) + \Delta(w) \rightarrow 0$ . Both are  $c$ -periodic, and a  $c$ -periodic function tending to 0 as  $w \rightarrow +\infty$  vanishes identically, its value at any  $w$  being its value at  $w + kc$  for every  $k$ . Hence  $\tilde{H} = H$ , and by uniform convergence the  $\hat{H}(m)$  are the Fourier coefficients of  $H$ .  $\square$

*Remark 6.4.* This is an instance of the de Bruijn–Mahler phenomenon: a base-change harmonic sum, Mellin transformed, picks up simple poles on a vertical line, and their residues assemble into an exactly periodic correction with Fourier coefficients in  $\Gamma$  and  $\zeta$ . De Bruijn found the first case of this in Mahler’s partition asymptotic [28]; Erdős and Richmond name it on p. 448 of [29]; Kato and McLeod’s §4 already exhibits the Gaussian-in-log decay that Section 5’s comparison asymptotic  $\varphi_0$  carries: their characterization (4.3) of the admissible asymptotics for  $y'(x) = ay(\lambda x) + by(x)$  is estimated there through  $\exp\{(n + \kappa) \log x - (\log x)^2/2 \log \lambda\}$  [30, §4, eqs. (4.3), (4.9a)], so the log-squared rate is generic across this class of rescaled equations rather than special to  $\varphi_0$ . Our contribution here is the closed form for the periodic correction itself, not the mechanism producing it.

The series for  $H'$  and  $H''$  converge absolutely and uniformly, since

$$|\Gamma(-i\omega_m)| = \sqrt{\pi/(\omega_m \sinh \pi\omega_m)}$$

decays exponentially in  $m$  while  $\zeta(1 - i\omega_m)$  grows only polynomially: for  $\omega \geq \alpha_H := 2\pi/c$  and  $N := \lfloor \omega \rfloor \geq 1$ , the classical Euler–Maclaurin formula for  $\zeta$ ,  $\zeta(s) = \sum_{n=1}^N n^{-s} + \frac{N^{1-s}}{s-1} - s \int_N^\infty \{u\} u^{-s-1} du$  (valid for  $\operatorname{Re} s > 0$ ,  $s \neq 1$ , integer  $N \geq 1$ , with  $\{u\} := u - \lfloor u \rfloor$ ), at  $s = 1 + i\omega$  gives, bounding each term in modulus ( $|N^{-i\omega}| = 1$ ,  $0 \leq \{u\} < 1$ ),

$$|\zeta(1 + i\omega)| \leq \sum_{n=1}^N \frac{1}{n} + \frac{1}{\omega} + \frac{\sqrt{1 + \omega^2}}{N} \leq 1 + \log N + \frac{1}{\alpha_H} + \frac{\sqrt{1 + \omega^2}}{N},$$

using  $\sum_{n=1}^N 1/n \leq 1 + \log N$  and  $\omega \geq \alpha_H$  in the last term. Since  $N \leq \omega$ ,  $\log N \leq \log \omega$ ; since  $N > \omega - 1$  and  $\omega \mapsto \omega/(\omega - 1)$  is decreasing for  $\omega > 1$ ,  $\sqrt{1 + \omega^2}/N < (\omega/(\omega - 1))\sqrt{1 + \omega^{-2}} \leq (\alpha_H/(\alpha_H - 1))\sqrt{1 + \alpha_H^{-2}}$ . With

$$C_H := 1 + \alpha_H^{-1} + \frac{\alpha_H}{\alpha_H - 1} \sqrt{1 + \alpha_H^{-2}},$$

this gives  $|\zeta(1 + i\omega)| \leq \log \omega + C_H \leq \log(\omega + 1) + C_H$  for  $\omega \geq \alpha_H$ , and  $\zeta(1 - i\omega) = \overline{\zeta(1 + i\omega)}$  has the same modulus, so  $|\zeta(1 - i\omega_m)| \leq \log(\omega_m + 1) + C_H$ . With  $q := e^{-\pi\alpha_H/2}$ ,  $A_H := c^{-1}\sqrt{3\pi/\alpha_H}$ , the bound  $|\Gamma(-i\omega_m)| \leq \sqrt{3\pi/\omega_m} e^{-\pi\omega_m/2}$  (valid for  $\omega_m \geq \alpha_H$ , since  $\sinh x \geq e^x/3$  there) together with the bound on  $\zeta(1 - i\omega_m)$  just proved give the majorant

$$|\hat{H}(m)| \leq A_H q^m m^{-1/2} (\log(\alpha_H m + 1) + C_H), \quad m \geq 1. \quad (9)$$

Since  $\sup_w |H^{(k)}(w)| \leq \sum_{m \neq 0} |\omega_m|^k |\hat{H}(m)| = 2 \sum_{m \geq 1} \omega_m^k |\hat{H}(m)|$ , summing the first six terms with rigorous ball enclosures and bounding the tail  $m > 6$  by (9) (the same closed-form geometric-times-polynomial tail sum used in the proof of Proposition 6.5 below) gives explicit, certified bounds:

$$\sup_w |H'(w)| \leq 0.0011977472315550332, \quad \sup_w |H''(w)| \leq 0.0068518962896650951. \quad (10)$$

**Proposition 6.5** (Non-constancy and certified oscillation).  *$H$  is not constant:  $\hat{H}(1) \neq 0$  follows from the classical zero-free theorem for  $\zeta$  on  $\operatorname{Re} s = 1$  alone, with no computation. The oscillation is quantified rigorously by an explicit interval-arithmetic certificate:*

$$-0.000377190280943987 < H(0) - H(\log(3/2)) < -0.000377190280943985,$$

and consequently  $\operatorname{osc}(H) := \sup H - \inf H \geq 3.7719 \times 10^{-4}$ ; evaluating  $H$  at two further, well-separated points certifies the stronger bound  $\operatorname{osc}(H) \geq 4.1874494771 \times 10^{-4}$ , and the leading Fourier mode together with a certified bound on the remaining modes gives a matching upper bound, the two giving

$$4.1874494771 \times 10^{-4} \leq \operatorname{osc}(H) \leq 4.187981 \times 10^{-4}.$$

*Proof.* Non-constancy first, unconditionally. By Proposition 6.3,  $\hat{H}(1) = -\frac{2^{i\omega_1}}{c}\Gamma(-i\omega_1)\zeta(1-i\omega_1)$ ,  $\omega_1 = 2\pi/c$ , a product of three factors, each nonzero:  $2^{i\omega_1} \neq 0$  trivially;  $\Gamma$  has no zeros anywhere, and  $-i\omega_1$  is not one of its poles ( $\omega_1 \neq 0$  is real, and  $\Gamma$ 's poles are only at the non-positive integers), so  $\Gamma(-i\omega_1) \neq 0$ ; and  $1-i\omega_1$  lies on the line  $\operatorname{Re} s = 1$  with  $\omega_1 \neq 0$ , where  $\zeta(1+it) \neq 0$  for every real  $t \neq 0$  by the classical zero-free theorem of Hadamard [31] and de la Vallée Poussin [32]<sup>1</sup>, so  $\zeta(1-i\omega_1) \neq 0$ . Hence  $\hat{H}(1) \neq 0$ , and a periodic function with a nonzero Fourier coefficient at  $m \neq 0$  cannot be constant.

For the quantitative bound, write  $g(b) = -b/2 + \log S(b/2)$ ,  $S(x) = \sinh(x)/x = \sum_{n \geq 0} x^{2n}/(2n+1)!$  (this is  $g$ 's defining formula rewritten around its removable singularity at 0, since  $e^{-b/2}(e^{b/2} - e^{-b/2}) = 1 - e^{-b}$ ). Termwise domination of  $6^n n! \leq (2n+1)!$ , which holds at  $n=0$  and propagates because  $(2n+3)(2n+2) \geq 6(n+1)$ , gives  $1 \leq S(x) \leq e^{x^2/6}$ , that is

$$0 \leq \log S(x) \leq \frac{x^2}{6}, \quad x \geq 0. \quad (11)$$

Since  $\sum_{j \geq 1} 3^{-j} = \frac{1}{2}$ , the  $-b/2$  halves of the terms of  $L$  sum in closed form, and (7) becomes, with  $s = e^w$ ,

$$H(w) = -\frac{s}{2} + \sum_{j \geq 1} \log S\left(\frac{s}{3^j}\right) - Q(w) + \sum_{k \geq 0} \log(1 - e^{-2 \cdot 3^k s}). \quad (12)$$

Truncate the first series at  $j = R$  and the second at  $k = M$ . By (11) the first tail satisfies

$$0 \leq \sum_{j > R} \log S\left(\frac{s}{3^j}\right) \leq \frac{s^2}{6} \sum_{j > R} 9^{-j} = \frac{s^2 9^{-R}}{48},$$

and, since  $0 \leq -\log(1-y) \leq y/(1-y)$  and  $e^{-2 \cdot 3^k s} = (e^{-2 \cdot 3^{M+1} s})^{3^{k-M-1}}$  for  $k > M$ , the second tail satisfies

$$0 \leq -\sum_{k > M} \log(1 - e^{-2 \cdot 3^k s}) \leq \frac{y}{(1-y)^2} \leq 4y, \quad y := e^{-2 \cdot 3^{M+1} s} \leq \frac{1}{2}.$$

Take  $R = 30$ ,  $M = 4$ . For  $s \leq 3$  (which covers every evaluation point used below, including the oscillation grid) the first tail is at most  $4.43 \times 10^{-30}$ , and for  $s \geq 1$  the second is at most  $4e^{-486} < 10^{-210}$ . The remaining 35 terms are elementary functions of  $s$ ; evaluated at 50 significant decimal digits, where the accumulated rounding stays below  $10^{-45}$ , they give

$$H(0) = -0.62713093305152597830\dots, \quad H(\log \frac{3}{2}) = -0.62675374277058199247\dots,$$

<sup>1</sup>The qualitative statement is what is used; no quantitative zero-free region is needed here.



whose difference is  $-0.00037719028094398583\dots$ , inside the stated interval; this first route is floating-point (50 significant digits, accumulated rounding bounded as stated) rather than interval arithmetic. The Fourier series of Proposition 6.3 gives the same difference in Arb ball arithmetic, rigorously: the  $m = 0$  term cancels, and at 250-bit working precision (ball radius  $< 10^{-70}$  throughout) the four nonzero modes needed are

$$\begin{aligned}\hat{H}(1) &= -1.020544273068431785 \times 10^{-4} - 2.332597606471452884 \times 10^{-5} i, \\ \hat{H}(2) &= -6.523276079851823231 \times 10^{-9} + 1.156720838500605666 \times 10^{-8} i, \\ \hat{H}(3) &= -1.799881020474424677 \times 10^{-12} + 6.648927223521397737 \times 10^{-13} i, \\ \hat{H}(4) &= -9.219524659678388767 \times 10^{-17} + 9.444517590655925766 \times 10^{-17} i,\end{aligned}$$

the four-mode sum

$$2 \operatorname{Re} \sum_{m=1}^4 \hat{H}(m) (1 - e^{i\omega_m \log(3/2)}) = -0.0003771902809439858148\dots$$

(ball radius  $< 10^{-70}$ ), matching the floating-point value above to 19 decimal places (16 significant digits). The majorant (9) gives  $\sum_{m>4} |\hat{H}(m)| \leq 9.41 \times 10^{-20}$  (the accompanying repository derives  $C_H$  by a slightly different route, reaching 2.182435 in place of 2.405136 and hence the tighter  $9.04 \times 10^{-20}$ ; either bound serves here), so the discarded modes move the four-mode sum by less than  $3.8 \times 10^{-19}$ , giving

$$H(0) - H(\log \tfrac{3}{2}) \in [-0.0003771902809439861948, -0.0003771902809439854348];$$

this route alone certifies the stated interval for  $H(0) - H(\log(3/2))$ .

For the stronger bound, let  $w_i = ic/N$ ,  $0 \leq i < N$ , with  $N = 2^{20}$ . A floating-point evaluation of the  $N$  grid values from Proposition 6.3, truncated at  $|m| \leq 10$  (the same majorant bounds each pointwise truncation error by  $2 \times 2.80 \times 10^{-43} = 5.60 \times 10^{-43}$ , combining the discarded modes  $m$  and  $-m$ ), locates a candidate maximum at  $i = 486746$  and a candidate minimum at  $i = 1011118$ ; this search is only a heuristic for finding two well-separated points and is not itself part of the certificate below. Re-evaluating  $H$  at exactly these two points at 250-bit working precision in Arb ball arithmetic, independently, from (12) and from the Fourier series (the two routes agree to about 30 digits, consistent with (12)'s own  $R = 30$  truncation tail bound above; a cross-check, not part of the certificate),

$$H(w_{486746}) = -0.6267174403736425786\dots, \quad H(w_{1011118}) = -0.6271361853213578093\dots,$$

so  $D := H(w_{486746}) - H(w_{1011118}) = 4.187449477152 \times 10^{-4}$ . The certified enclosure uses only the Fourier route: the Arb ball rounding at 250-bit precision is  $< 10^{-70}$  per evaluation, negligible next to the majorant truncation error already bounded above,  $5.60 \times 10^{-43}$  per point, giving  $D$  to radius  $< 1.2 \times 10^{-42}$  (twice that per-point bound). Since  $w_{486746}$  and  $w_{1011118}$  are two specific, fixed points of the domain,  $H(w_{486746}) \leq \sup H$  and  $H(w_{1011118}) \geq \inf H$  hold trivially, giving  $D \leq \operatorname{osc}(H)$ : this bound is unconditional, independent of whether the floating-point search above actually located the true argmax and argmin, since it uses only that the two chosen points lie in the domain.

For the matching upper bound, no search is needed. Write  $H = \hat{H}(0) + 2 \operatorname{Re}(\hat{H}(1)e^{i\omega_1 w}) + \Xi(w)$ ,  $\Xi(w) := 2 \sum_{m \geq 2} \operatorname{Re}(\hat{H}(m)e^{i\omega_m w})$ , so  $|\Xi(w)| \leq 2 \sum_{m \geq 2} |\hat{H}(m)|$  for every  $w$ . The one-mode term  $2 \operatorname{Re}(\hat{H}(1)e^{i\omega_1 w})$  ranges, as  $w$  varies, over an interval of width exactly  $4|\hat{H}(1)|$ , so  $\operatorname{osc}(H) \leq 4|\hat{H}(1)| +$

$4 \sum_{m \geq 2} |\hat{H}(m)|$ . Using the already-certified  $\hat{H}(2)$ ,  $\hat{H}(3)$ ,  $\hat{H}(4)$  from above and the majorant tail  $\sum_{m > 4} |\hat{H}(m)| \leq 9.41 \times 10^{-20}$ ,

$$\sum_{m \geq 2} |\hat{H}(m)| \leq |\hat{H}(2)| + |\hat{H}(3)| + |\hat{H}(4)| + 9.41 \times 10^{-20} \leq 1.3282 \times 10^{-8},$$

and  $4|\hat{H}(1)| \leq 4.187449304 \times 10^{-4}$ , giving  $\text{osc}(H) \leq 4.187981 \times 10^{-4}$ , rigorously and unconditionally: this bound uses only the already-certified Fourier coefficients, not the grid search.  $\square$

## 7 A uniform saddlepoint approximation

Define, for  $s > 0$ ,

$$\varphi_{\text{sp}}(t(s)) := \frac{\exp(K(s) + st(s))}{\sqrt{2\pi K''(s)}}.$$

This section proves that  $\varphi(t(s))/\varphi_{\text{sp}}(t(s)) \rightarrow 1$  as  $s \rightarrow \infty$ , uniformly over  $\rho := 2s/3^N \in [1, 3)$  (defined below; this parameter records the same phase,  $w \bmod c$ , tracked through  $H$  in Sections 9 and 10). That uniformity is what lets the two theorems of Section 10 depend on different scopes rather than on different mathematics.

For  $n \geq 2$ , define the derivative-normalized functions  $\kappa_n(w) = (-1)^n w^n g^{(n)}(w)$  on  $\{\text{Re } w > 0\}$ ;  $\kappa_n(b)$ ,  $b > 0$ , is  $b^n$  times the  $n$ -th cumulant of the random variable with density  $b e^{-bw}/(1 - e^{-b})$  on  $[0, 1]$ . Writing  $x = w/2$ ,

$$\kappa_2(w) = 1 - \frac{x^2}{\sinh^2 x}, \quad \kappa_3(w) = 2 - \frac{2x^3 \cosh x}{\sinh^3 x}, \quad \kappa_4(w) = 6 + \frac{2x^4(1 - 3 \coth^2 x)}{\sinh^2 x},$$

and from the Bernoulli expansion  $g(w) + w/2 = \sum_{k \geq 1} B_{2k} w^{2k}/(2k(2k)!)$  on  $|w| < 2\pi$ ,

$$\kappa_n(w) = (-1)^n \sum_{k \geq \lceil n/2 \rceil} \frac{B_{2k} w^{2k}}{2k(2k-n)!}, \quad n \geq 2,$$

so that  $\kappa_n(w) = O(w^2)$  as  $w \rightarrow 0$  for every  $n \geq 2$ .

Write  $s = \rho \cdot 3^N/2$  for the unique  $N = \lfloor \log_3(2s) \rfloor \in \mathbb{Z}$  and  $\rho \in [1, 3)$ , and  $b_j = 2s/3^j$ . Then  $\{b_j : j \geq 1\} = \{\rho \cdot 3^m : m \leq N-1\}$ ; for  $N \geq 1$  (the only range used from here on, corresponding to  $s \geq 3/2$ ), exactly  $N$  of these, the smallest being  $\rho \geq 1$ , are  $\geq 1$ , and the rest form a geometric tail below 1. This single structural fact is the source of every phase-independent constant below.

**Lemma 7.1** (Boundedness off the real axis). *For  $w = b(1 + iu)$ ,  $b > 0$ ,  $|u| \leq 1$ , with  $e(r) = r^2/\sinh^2 r$  and  $f(r) = r^3 \cosh r/\sinh^3 r$ ,*

$$|\kappa_2(w)| \leq 1 + 2e(b/2), \quad |\kappa_3(w)| \leq 2 + 2 \cdot 2^{3/2} f(b/2).$$

*Both bounds are finite over the whole sector  $|\text{Im } w| \leq \text{Re } w$ :  $\sup |\kappa_2| \leq 3$ ,  $\sup |\kappa_3| \leq 2 + 2^{5/2} < 7.657$ . In addition, for  $|w| \leq 2$ ,  $|\kappa_2(w)| \leq 0.114|w|^2$  and  $|\kappa_3(w)| \leq 0.0119|w|^4$ .*

*Proof.*  $|\sinh z|^2 = \sinh^2(\text{Re } z) + \sin^2(\text{Im } z) \geq \sinh^2(\text{Re } z)$  and  $|\cosh z|^2 \leq \cosh^2(\text{Re } z)$ ; the first gives  $|\sinh(w/2)| \geq \sinh(b/2) > 0$ , excluding the poles of  $\coth$ , since  $\text{Re}(w/2) = b/2 > 0$ . This gives the sector bounds directly. Monotonicity of  $e$  is immediate from the increasing power series of  $\sinh(r)/r$ . For  $f$ , set  $u = 2r$ ; a direct computation gives  $2r \sinh(r) \cosh(r) (\log f)'(r) = F(u)$  with  $F(u) := 3 \sinh u - u \cosh u - 2u$ , so  $(\log f)'(r)$  has the sign of  $F(2r)$ . Since  $F(0) = F'(0) = F''(0) = 0$

and  $F'''(u) = -u \sinh u < 0$  for  $u > 0$ , integrating three times from 0 gives  $F(u) < 0$  for all  $u > 0$ , hence  $(\log f)'(r) < 0$  for  $r > 0$  and  $f$  is strictly decreasing. (Lazarević's inequality,  $(\sinh r/r)^3 > \cosh r$ , separately gives  $f < 1$ .) The local bounds at  $|w| \leq 2$  follow by applying the maximum-modulus principle to  $\kappa_2(w)/w^2$  and  $\kappa_3(w)/w^4$  (analytic on  $|w| < 2\pi$  by the Bernoulli expansion): bound each on  $|w| = 2$  by the triangle inequality, using  $|B_{2k}| = 2(2k)! \zeta(2k)/(2\pi)^{2k}$  and  $\zeta(2k) \leq \zeta(2)$ , respectively  $\zeta(4)$ .  $\square$

**Lemma 7.2** (Variance). *For all  $\rho \in [1, 3)$ ,  $N \geq 1$ , with  $V := s^2 K''(s) = \sum_j \kappa_2(b_j)$ ,*

$$N - A_V \leq V \leq N + 0.1283, \quad A_V := \sum_{m \geq 0} e(3^m/2) = 1.4269413069 \dots$$

*Proof.* Split  $V = \sum_{m=0}^{N-1} \kappa_2(\rho 3^m) + \sum_{m \geq 0} \kappa_2(\rho 3^m)$ , using  $\kappa_2(b) = 1 - e(b/2) \in (0, 1)$ . For the lower bound, drop the second (non-negative) sum and use  $e(\rho 3^m/2) \leq e(3^m/2)$  ( $e$  decreasing,  $\rho \geq 1$ ) in the first, giving  $\sum_{m=0}^{N-1} \kappa_2(\rho 3^m) \geq N - A_V$ . For the upper bound, the first sum is  $< N$  term by term, and the second sum, where  $\rho 3^m < 1$ , is bounded by Lemma 7.1's local estimate:  $\sum_{m \geq 0} \kappa_2(\rho 3^m) \leq 0.114 \rho^2 \sum_{m \geq 0} 9^m = 0.114 \rho^2 / 8 \leq 0.1283$  for  $\rho < 3$ .  $\square$

**Lemma 7.3** (Tail bound and existence of the density). *If  $b_0 \geq 1$  and  $b_k = 3^k b_0$ , then*

$$\sum_{k \geq 0} \log \coth(b_k/2) \leq 3e^{-b_0}.$$

*Consequently, with  $\Lambda(y) := K(s(1 + iy)) - K(s) + isy t(s)$ ,*

$$|e^{\Lambda(y)}| \leq e^{3/e} (1 + y^2)^{-N/2} \leq 3.0152 (1 + y^2)^{-N/2}$$

*for every real  $y$ . In particular  $X$  has a continuous density, the tilted characteristic function is integrable for  $N \geq 3$ , and Fourier inversion at  $x = t(s)$  gives*

$$R(s) := \frac{\varphi(t(s))}{\varphi_{\text{sp}}(t(s))} = \sqrt{\frac{V}{2\pi}} \int_{\mathbb{R}} e^{\Lambda(y)} dy, \quad (13)$$

*a real quantity, since  $\Lambda(-y) = \overline{\Lambda(y)}$ .*

*Proof.*  $\log \coth(x/2) = 2 \operatorname{artanh}(e^{-x}) \leq 2e^{-x}/(1 - e^{-2x})$ . Write  $q := e^{-b_0} \in (0, e^{-1}]$ , so  $e^{-b_k} = q^{3^k}$  and the bound reads  $\log \coth(b_k/2) \leq 2q^{3^k}/(1 - q^{2 \cdot 3^k})$ . Since  $3^k \geq 1$  for  $k \geq 0$ ,  $q^{2 \cdot 3^k} \leq q^2$  term by term, so the denominator is bounded below by  $1 - q^2$  uniformly in  $k$ :

$$\sum_{k \geq 0} \log \coth(b_k/2) \leq \frac{2}{1 - q^2} \sum_{k \geq 0} q^{3^k}.$$

Since  $3^k \geq 2k+1$  for every  $k \geq 0$  and  $0 < q < 1$ ,  $q^{3^k} \leq q^{2k+1}$ , so  $\sum_{k \geq 0} q^{3^k} \leq q \sum_{k \geq 0} q^{2k} = q/(1 - q^2)$ , giving  $\sum_{k \geq 0} \log \coth(b_k/2) \leq 2q/(1 - q^2)^2$ . For  $b_0 \geq 1$ ,  $q \leq e^{-1} < 0.4283$ , so  $(1 - q^2)^2 > 2/3$  and  $2q/(1 - q^2)^2 < 3q = 3e^{-b_0}$ .  $X$ 's density is continuous because the sum of the first two summands already has a continuous, compactly supported density and convolution preserves continuity. For  $\Lambda$ :  $t(s) = -K'(s)$  cancels the linear term of the tilted transform exactly, and each remaining factor  $e^{g(b_j(1+iy)) - g(b_j)}$  has modulus at most 1 (a tilted characteristic function), with modulus at most  $\coth(b_j/2)/\sqrt{1 + y^2}$  on the  $N$  indices where  $b_j \geq 1$ ; the tail estimate applies to exactly that geometric set, with  $b_0 = \rho \geq 1$ .  $\square$

**Lemma 7.4** (Central expansion). *With  $h(u) = g(b(1+iu))$ , Taylor's theorem in integral form (the Lagrange form is invalid for a complex-valued function of a real variable) gives*

$$h(y) = h(0) + h'(0)y + h''(0)\frac{y^2}{2} + \frac{1}{2} \int_0^y (y-u)^2 h'''(u) du.$$

Consequently, writing  $h_j(u) := g(b_j(1+iu))$  for each  $j \geq 1$ , for  $\theta \in (0, 1]$ ,

$$S := \sum_j \sup_{|u| \leq \theta} |h_j'''(u)| \leq 2N + 10.559,$$

and hence  $|\Lambda(y) + Vy^2/2| \leq B|y|^3$  for  $|y| \leq \theta$ , with  $B := S/6 \leq (2N + 10.559)/6$ .

*Proof.* Write  $w_j := b_j(1+iu)$ . The chain rule and the definition of  $\kappa_3$  give  $h_j'''(u) = i\kappa_3(w_j)/(1+iu)^3$ , so

$$|h_j'''(u)| = \frac{|\kappa_3(w_j)|}{(1+u^2)^{3/2}}.$$

Split the index set as in Section 7. For the  $N$  indices with  $b_j = \rho 3^m \geq 1$ ,  $0 \leq m \leq N-1$ , drop the denominator and use Lemma 7.1's sector bound, legitimate since  $|u| \leq \theta \leq 1$ :

$$|h_j'''(u)| \leq |\kappa_3(w_j)| \leq 2 + 2^{5/2}f(b_j/2) \leq 2 + 2^{5/2}f(3^m/2),$$

the last step by monotonicity of  $f$  and  $\rho \geq 1$ . Summing over  $m \geq 0$  bounds this block by  $2N + 2^{5/2}\Sigma$ ,  $\Sigma := \sum_{m \geq 0} f(3^m/2)$ . The first three terms of  $\Sigma$  are  $0.99614738\dots$ ,  $0.82241066\dots$ ,  $0.04500508\dots$ ; for  $r \geq 13$  one has  $f(r) \leq 4.001r^3e^{-2r}$ , so the remaining terms sum to less than  $2 \times 10^{-8}$ , and  $2^{5/2}\Sigma < 10.541906$ .

For the remaining indices  $b_j = \rho 3^{-n}$ ,  $n \geq 1$ , one has  $b_j < 1$  and  $|w_j| = b_j\sqrt{1+u^2} \leq b_j\sqrt{2} < 2$ , so Lemma 7.1's local bound applies and, keeping the denominator this time,

$$|h_j'''(u)| \leq \frac{0.0119|w_j|^4}{(1+u^2)^{3/2}} = 0.0119b_j^4(1+u^2)^{1/2} \leq 0.0119\sqrt{2}b_j^4.$$

Summing,  $\sum_{n \geq 1} (\rho 3^{-n})^4 = \rho^4/80 < 81/80$ , so this block is below  $0.0119\sqrt{2} \cdot 81/80 < 0.01704$ . Together,  $S < 2N + 10.541906 + 0.01704 < 2N + 10.559$ .

For the remainder bound,  $\sum_j h_j(y) = K(s(1+iy))$ ,  $\sum_j h_j'(0) = isK'(s) = -ist(s)$  and  $\sum_j h_j''(0) = -s^2K''(s) = -V$ , so summing the integral-form Taylor expansion over  $j$  gives

$$\Lambda(y) + \frac{Vy^2}{2} = \sum_j \frac{1}{2} \int_0^y (y-u)^2 h_j'''(u) du,$$

whose modulus is at most  $S|y|^3/6 = B|y|^3$  for  $|y| \leq \theta$ . □

**Theorem 7.5** (Uniform saddlepoint asymptotic). *For  $N \geq 19$ , uniformly over  $\rho \in [1, 3)$ ,  $|R(s) - 1| \leq E(N) := e_1(N) + e_2(N) + e_3(N)$ , with  $\theta_N := N^{-1/3}$ ,  $V_{\text{lo}} := N - A_V$ ,  $V_{\text{up}} := N + 0.1283$ ,  $B := (2N + 10.559)/6$ ,*

$$\begin{aligned} e_1(N) &:= \frac{4e^{B\theta_N^3} B}{\sqrt{2\pi} V_{\text{lo}}^{3/2}}, & e_2(N) &:= \frac{2}{\sqrt{2\pi}} \cdot \frac{e^{-\theta_N^2 V_{\text{lo}}/2}}{\theta_N \sqrt{V_{\text{lo}}}}, \\ e_3(N) &:= 2e^{3/e} \sqrt{\frac{V_{\text{up}}}{2\pi}} \cdot \frac{(1 + \theta_N^2)^{-(N-2)/2}}{\theta_N(N-2)}. \end{aligned} \tag{14}$$

$\sqrt{N} E(N) \rightarrow \frac{4}{3} e^{1/3} (2\pi)^{-1/2} = 0.742358\dots$  as  $N \rightarrow \infty$  (proved below, from  $e_1$  alone:  $e_2$  and  $e_3$  vanish faster than any power of  $N$ ). Consequently  $\varphi(t) = \varphi_{\text{sp}}(t)(1 + o(1))$  as  $t \rightarrow 0^+$ , with no restriction on the sequence of  $t$ 's.

*Remark 7.6.* The estimate  $|R(s) - 1| \leq E(N)$  becomes informative once  $E(N) < 1$ , which holds at  $N = 19$  (see the proof:  $E(18) = 1.00170\dots \geq 1$ ,  $E(19) = 0.95673\dots < 1$ ); this is the threshold kept as the theorem's hypothesis (whether some smaller  $N$  also gives  $E(N) < 1$  is immaterial to what follows and is not checked).  $E$  is also strictly decreasing and satisfies  $E(N) \leq 4.18 N^{-1/2}$  throughout  $19 \leq N \leq 5000$  (checked directly in the accompanying repository), though only  $E(N) \rightarrow 0$  as  $N \rightarrow \infty$  is used below.

*Proof.* Split (13) at  $|y| = \theta_N$ . On  $|y| \leq \theta_N$ , Lemma 7.4 bounds the cubic remainder  $|\Lambda(y) + Vy^2/2|$  by  $B\theta_N^3$ , which stays bounded (not  $o(1)$ : this is exactly why  $e_1$  carries the factor  $e^{B\theta_N^3}$  rather than vanishing) as  $N \rightarrow \infty$ , since  $B = O(N)$  and  $\theta_N^3 = N^{-1}$ . Write  $C(y) := e^{\Lambda(y)} - e^{-Vy^2/2}$  and factor the Gaussian out before estimating,  $e^{\Lambda(y)} = e^{-Vy^2/2} e^{z(y)}$  with  $z(y) := \Lambda(y) + Vy^2/2$ , so that  $C(y) = e^{-Vy^2/2}(e^{z(y)} - 1)$ . Since  $|z(y)| \leq B|y|^3 \leq B\theta_N^3$  on  $|y| \leq \theta_N$ , the elementary  $|e^z - 1| \leq |z|e^{|z|}$  gives

$$|C(y)| \leq e^{-Vy^2/2} B|y|^3 e^{B\theta_N^3} \quad (|y| \leq \theta_N),$$

the Gaussian factor retained. Integrating that inequality,  $\int_{\mathbb{R}} e^{-Vy^2/2} |y|^3 dy = 4/V^2$ , so

$$\sqrt{V/2\pi} \int_{|y| \leq \theta_N} |C(y)| dy \leq \frac{4Be^{B\theta_N^3}}{\sqrt{2\pi} V^{3/2}},$$

which is decreasing in  $V$ ; using  $V \geq V_{\text{lo}}$  from Lemma 7.2 gives the central Taylor-remainder contribution  $e_1$ . Since  $\sqrt{V/2\pi} \int_{\mathbb{R}} e^{-Vy^2/2} dy = 1$ , the same splitting leaves exactly two further terms. The truncated Gaussian tail  $\sqrt{V/2\pi} \cdot 2 \int_{\theta_N}^{\infty} e^{-Vy^2/2} dy$ , substituting  $v = y\sqrt{V}$  to get  $(2/\sqrt{2\pi}) \int_{c_0}^{\infty} e^{-v^2/2} dv$ ,  $c_0 = \theta_N\sqrt{V}$  (a local constant, distinct from  $c := \log 3$ ), and bounding via  $\int_{c_0}^{\infty} e^{-v^2/2} dv \leq e^{-c_0^2/2}/c_0$  at  $V = V_{\text{lo}}$ , gives  $e_2$ . On  $|y| > \theta_N$ , Lemma 7.3's bound  $|e^{\Lambda(y)}| \leq e^{3/e}(1+y^2)^{-N/2}$  integrates, via  $\int_u^{\infty} (1+y^2)^{-N/2} dy \leq (1+u^2)^{-(N-2)/2}/(u(N-2))$  at  $u = \theta_N$  (using  $V \leq V_{\text{up}}$  for the leading  $\sqrt{V/2\pi}$  factor in (13)), to give  $e_3$ ; this needs  $\theta_N \gg N^{-1/2}$ , satisfied since  $\frac{1}{3} < \frac{1}{2}$ . Every constant entering  $E$  (via  $A_V$ , Lemma 7.2, and the  $3.0152 \geq e^{3/e}$  of Lemma 7.3) is independent of  $\rho$ , by one of three routes: it is a supremum over  $b > 0$ ; or a sum over one complete geometric orbit bounded using  $\rho \geq 1$ ; or a bound at the other endpoint,  $\rho \rightarrow 3^-$ , which is where  $V_{\text{up}}$ 's 0.1283 and  $B$ 's 0.01704 come from. Uniformity in  $\rho$  is the theorem's whole content, so it is worth saying which constant is which rather than covering all of them with one clause. Directly evaluating (14) gives  $E(18) = 1.00170\dots \geq 1$  and  $E(19) = 0.95673\dots < 1$ , matching the remark above.

For the asymptotic rate,  $e_1, e_2, e_3$  separate on very different scales as  $N \rightarrow \infty$ . Since  $\theta_N^3 = 1/N$ ,  $B\theta_N^3 \rightarrow 1/3$  and  $B/N \rightarrow 1/3$ ,  $V_{\text{lo}}/N, V_{\text{up}}/N \rightarrow 1$ , so

$$\sqrt{N} e_1(N) = \frac{4e^{B\theta_N^3}}{\sqrt{2\pi}} \cdot \frac{\sqrt{N} B}{V_{\text{lo}}^{3/2}} \rightarrow \frac{4e^{1/3}}{\sqrt{2\pi}} \cdot \frac{1}{3} = \frac{4}{3} e^{1/3} (2\pi)^{-1/2}.$$

For  $e_2$  and  $e_3$ ,  $\theta_N^2 V_{\text{lo}} \sim N^{1/3} \rightarrow \infty$ , so both carry a factor  $e^{-N^{1/3}/2}(1 + o(1))$  (for  $e_3$ , via  $(1 + \theta_N^2)^{-(N-2)/2} = \exp(-\frac{N-2}{2}\theta_N^2(1 + o(1)))$ ), against polynomial prefactors of order  $N^{-1/6}$ ; a factor decaying like  $e^{-N^{1/3}/2}$  beats every power of  $N$ , so  $N^k e_2(N), N^k e_3(N) \rightarrow 0$  for every  $k$ . Hence  $\sqrt{N} E(N) = \sqrt{N} e_1(N) + o(1) \rightarrow \frac{4}{3} e^{1/3} (2\pi)^{-1/2}$ , and in particular  $E(N) \rightarrow 0$ . Since  $t(s)$  ranges bijectively over  $(0, 1/2)$ , this is exactly the statement  $\varphi(t) = \varphi_{\text{sp}}(t)(1 + o(1))$  as  $t \rightarrow 0^+$ .  $\square$

*Remark 7.7.* The rate is not sharp, and a sharper one is plausible but not established here. Carrying the central expansion to fourth order, using that the odd cubic term integrates to zero against the Gaussian core, and using the limits  $V_3 := \sum_j \kappa_3(b_j) = 2N + O(1)$ ,  $V_4 := \sum_j \kappa_4(b_j) = 6N + O(1)$  (the third and fourth cumulant sums, by the same argument as Lemma 7.2) suggests the Edgeworth correction  $R(s) = 1 - \frac{1}{12N} + O(N^{-3/2})$ , matching the leading term of the classical Stirling correction for a  $\text{Gamma}(N, 1)$  law: after tilting, the  $b_j \geq 1$  summands of  $sX = \sum_j b_j U_j$  behave asymptotically like independent  $\text{Exp}(1)$  variables, so  $sX$  under its tilt is asymptotically  $\text{Gamma}(N, 1)$ -distributed, and  $\rho$  enters only through  $O(1)$  shifts of the cumulants, at order  $N^{-2}$ . We do not need this sharper rate and do not prove the remainder term here; Theorem 7.5's qualitative  $o(1)$  is all that Section 10 uses.

## 8 The envelope lemma

Theorem 7.5 compares  $\varphi$  to  $\varphi_{\text{sp}}$ , a quantity built from the full transform  $K = Q + H + \Delta$ . It remains to compare  $\varphi_{\text{sp}}$  to a smooth auxiliary system built from  $Q$  alone, so that the periodic part  $H$  can be extracted explicitly. Write  $t = e^{-\tau}$ ,  $g_\tau(w) = L(w) + e^{w-\tau}$ ,  $g_{0,\tau}(w) = Q(w) + e^{w-\tau}$ . The true and smooth systems each have their own analogue of  $-K'$ : write  $B_{\text{tr}}(w) := e^w t(e^w) = -L'(w)$  and  $B_{\text{sm}}(w) := -Q'(w) = w/c - a$ .

**Lemma 8.1** (True and smooth saddles). *Let  $\Phi(w) = w - \log B_{\text{tr}}(w)$  and  $\Phi_0(w) = w - \log B_{\text{sm}}(w)$ , the latter defined for  $w > ca$  where  $B_{\text{sm}}(w) > 0$ , and set  $\tau_0 := 1 + ca + \log c = 0.9502067913\dots$ , so  $\tau_0 > \log 2$ .*

*For  $\tau > \log 2$ ,  $g_\tau$  has a unique global minimizer  $w^*(\tau)$ , at which  $g_\tau''(w^*) = V(r^*) > 0$ ,  $r^* = e^{w^*}$ ; it is characterized by  $\tau = \Phi(w^*)$ , and  $\Phi$  is a strictly increasing bijection  $\mathbb{R} \rightarrow (\log 2, \infty)$ . For  $\tau \leq \log 2$  there is no critical point at all. For  $\tau > \tau_0$ ,  $g_{0,\tau}$  has a unique local minimizer  $w_0(\tau)$ , characterized by  $\tau = \Phi_0(w_0)$  together with  $B_0 := B_{\text{sm}}(w_0) = e^{w_0-\tau} > 1/c$ ;  $\Phi_0$  restricted to  $(ca+1, \infty)$  is a strictly increasing bijection onto  $(\tau_0, \infty)$ , and  $B_0$  increases from  $1/c$  to  $\infty$  along it. For  $\tau \leq \tau_0$  there is no such minimizer.*

*Once  $B_0 \geq 5$  (equivalently  $\tau \geq 3.73978\dots$ ),  $|w^* - w_0| \leq 0.00652/B_0$ . Also  $w_0(\tau + c) - w_0(\tau) = c + O(1/\tau)$  as  $\tau \rightarrow \infty$ .*

*Proof.* Substituting  $s = e^w$  and  $t = e^{-\tau}$  turns  $g_\tau$  into  $K(s) + st$ , strictly convex in  $s > 0$  because  $K'' > 0$ , with derivative  $K'(s) + t$  vanishing exactly where  $t = t(s)$ . As  $t(\cdot)$  decreases strictly from  $1/2$  to  $0$ , that happens for exactly one  $s$  when  $t \in (0, 1/2)$ , that is when  $\tau > \log 2$ , and for no  $s$  when  $t \geq 1/2$ . Since  $w \mapsto e^w$  is a bijection  $\mathbb{R} \rightarrow (0, \infty)$ ,  $w^* = \log s$  is the unique global minimizer of  $g_\tau$ , and  $\Phi$  inverts  $\tau \mapsto w^*$ . Differentiating twice,  $g_\tau''(w) = e^w(K'(e^w) + t) + e^{2w}K''(e^w)$ , whose first term vanishes at  $w^*$ , leaving  $g_\tau''(w^*) = V(r^*) > 0$ .

For the smooth system,  $g_{0,\tau}'(w) = e^{w-\tau} - B_{\text{sm}}(w)$  and  $g_{0,\tau}''(w) = e^{w-\tau} - 1/c$ , so a critical point is a local minimizer exactly when  $B_0 > 1/c$ . Also  $\Phi_0'(w) = 1 - 1/(cB_{\text{sm}}(w))$ , positive exactly where  $B_{\text{sm}}(w) > 1/c$ , that is on  $w > ca + 1$ ; there  $\Phi_0$  increases from  $\Phi_0(ca + 1) = ca + 1 + \log c = \tau_0$  to  $\infty$ . This gives the stated existence, uniqueness and range.

Now take  $B_0 \geq 5$ . The decomposition  $L = Q + H + \Delta$  evaluates the true gradient at the smooth saddle exactly:  $Q'(w_0) = -B_0 = -e^{w_0-\tau}$ , so the leading terms cancel and

$$g_\tau'(w_0) = L'(w_0) + e^{w_0-\tau} = H'(w_0) + \Delta'(w_0), \quad |g_\tau'(w_0)| \leq \varepsilon_1 := \sup |H'| + \sup_{[w_0-1, w_0+1]} |\Delta'|. \quad (15)$$

On  $[w_0 - 1, w_0 + 1]$  write  $g_\tau''(w) = e^{w-\tau} - 1/c + H''(w) + \Delta''(w)$ . Here  $e^{w-\tau} = B_0 e^{w-w_0}$  ranges over



$[B_0/e, B_0e]$ , so  $g''_\tau$  admits no bound independent of  $\tau$ ; it is of exact order  $B_0$ ,

$$\frac{B_0}{e} - \eta_0 \leq g''_\tau(w) \leq B_0e, \quad \eta_0 := \frac{1}{c} + \sup |H''| + \sup_{[w_0-1, w_0+1]} |\Delta''|, \quad (16)$$

the upper bound because  $-1/c + \sup |H''| + \sup |\Delta''| < 0$ . Since  $w_0 = c(B_0 + a)$ ,  $B_0 \geq 5$  forces  $w_0 \geq 5.3492$  and so  $w \geq 4.3492$  on the interval, where  $|\Delta'|, |\Delta''| \leq 10^{-60}$ : with  $b := 2e^w \geq 2e^{4.3492} > 154$  and  $x_k := 3^k b$ , differentiating  $\Delta(w) = -\sum_{k \geq 0} \log(1 - e^{-x_k})$  termwise (justified as in the proof of Theorem 6.1) gives  $\Delta'(w) = -\sum_{k \geq 0} x_k/(e^{x_k} - 1)$  and, applying  $|d/dx[x/(e^x - 1)]| \leq 8xe^{-x}$  for  $x \geq 1$  (elementary, since  $e^x - 1 \geq e^x/2$  there) together with  $dx_k/dw = x_k$ ,  $\Delta''(w) = -\sum_{k \geq 0} x_k \cdot \frac{d}{dx} \left[ \frac{x}{e^x - 1} \right]_{x=x_k}$ , so  $|\Delta''(w)| \leq \sum_{k \geq 0} 8x_k^2 e^{-x_k}$ ; bounding both sums by their dominant  $k = 0$  term via  $x_k - b \geq 2kb$  (from  $3^k - 1 \geq 2k$ ) gives  $|\Delta'(w)| \leq 4be^{-b}$  and  $|\Delta''(w)| \leq 16b^2 e^{-b}$ , both well under  $10^{-60}$  at  $b > 154$ . With (10) this makes  $\varepsilon_1 \leq 0.0011978$  and  $\eta_0 \leq 0.9170912$ , whence  $2e\eta_0 \leq 4.9859 \leq 5 \leq B_0$ , that is  $\eta_0 \leq B_0/(2e)$ , and (16) improves to  $g''_\tau \geq B_0/(2e) > 0$  throughout the interval.

Put  $h_0 := 2e\varepsilon_1/B_0 \leq 2e \cdot 0.0011978/5 < 0.0014$ . For  $h \in (h_0, 1]$  the mean value theorem applied on  $[w_0 - 1, w_0 + 1]$  gives  $g'_\tau(w_0 + h) \geq g'_\tau(w_0) + hB_0/(2e) \geq -\varepsilon_1 + hB_0/(2e) > 0$  and, symmetrically,  $g'_\tau(w_0 - h) \leq \varepsilon_1 - hB_0/(2e) < 0$ . As  $w^*$  is the only zero of  $g'_\tau$ , it lies in  $(w_0 - h, w_0 + h)$  for every such  $h$ , so  $|w^* - w_0| \leq h_0 = 2e\varepsilon_1/B_0 \leq 0.00652/B_0$ .

For the phase drift,  $w_0 = c(B_0 + a)$  and  $\tau = w_0 - \log B_0$  give  $dw_0/d\tau = (1 - 1/(cB_0))^{-1} = 1 + O(1/B_0)$  along the curve  $\tau \mapsto w_0(\tau)$ ; since  $B_0 \rightarrow \infty$  as  $\tau \rightarrow \infty$  (via  $B_0 \sim \tau/c$ ), integrating over one period gives  $w_0(\tau + c) - w_0(\tau) = \int_\tau^{\tau+c} (1 + O(1/B_0(u))) du = c + O(1/\tau)$ .  $\square$

**Proposition 8.2** (Envelope estimate). *With  $S(\tau) := \log \varphi_{\text{sp}}(t) = g_\tau(w^*) + w^* - \frac{1}{2} \log(2\pi V(r^*))$  and  $P(\tau) := Q(w_0) + B_0 + w_0 - \frac{1}{2} \log(2\pi(B_0 - \frac{1}{c}))$ ,*

$$|S(\tau) - P(\tau) - H(w_0(\tau))| = O(1/\tau) \quad (\tau \rightarrow \infty),$$

*uniformly in the phase of  $\tau$  modulo  $c$ .*

*Proof.* Start from the identity. Since  $g_{0,\tau}(w_0) = Q(w_0) + e^{w_0-\tau} = Q(w_0) + B_0$ , the definition of  $P$  reads  $P(\tau) = g_{0,\tau}(w_0) + w_0 - \frac{1}{2} \log(2\pi(B_0 - \frac{1}{c}))$ , so

$$S(\tau) - P(\tau) = [g_\tau(w^*) - g_{0,\tau}(w_0)] + (w^* - w_0) - \frac{1}{2} \log \frac{V(r^*)}{B_0 - \frac{1}{c}}.$$

Theorem 6.1 gives  $L = Q + H + \Delta$ , hence  $g_\tau = g_{0,\tau} + H + \Delta$  pointwise, and in particular  $g_\tau(w^*) = g_{0,\tau}(w^*) + H(w^*) + \Delta(w^*)$ . Substituting and subtracting  $H(w_0)$ ,

$$\begin{aligned} S(\tau) - P(\tau) - H(w_0) &= \underbrace{[g_{0,\tau}(w^*) - g_{0,\tau}(w_0)]}_{T_1} + \underbrace{[H(w^*) - H(w_0)]}_{T_2} + \underbrace{(w^* - w_0)}_{T_3} \\ &\quad - \underbrace{\frac{1}{2} \log \frac{V(r^*)}{B_0 - \frac{1}{c}}}_{T_4} + \underbrace{\Delta(w^*)}_{T_5}. \end{aligned} \quad (17)$$

This is exact for every  $\tau$  with  $B_0 \geq 5$ . The five terms are bounded in turn.

$T_1$  is the envelope gap of the smooth system. Taylor about  $w_0$ , where  $g'_{0,\tau}$  vanishes, with  $g''_{0,\tau}(w) = e^{w-\tau} - 1/c \leq B_0e$  on  $[w_0 - 1, w_0 + 1]$  and  $|w^* - w_0| \leq 0.00652/B_0$  from Lemma 8.1, gives  $0 \leq T_1 \leq \frac{1}{2} B_0e (0.00652/B_0)^2 < 5.78 \times 10^{-5}/B_0$ .

$T_2$  is the periodic-part difference:  $|T_2| \leq \sup |H'| \cdot |w^* - w_0| < 7.82 \times 10^{-6}/B_0$  by (10).

$T_3$  is the saddle-location shift itself,  $|T_3| \leq 0.00652/B_0$ , again from Lemma 8.1.

$T_4$  compares the two normalizations. By  $V(r^*) = g_\tau''(w^*)$  and the expression for  $g_\tau''$  used in Lemma 8.1,

$$V(r^*) - \left(B_0 - \frac{1}{c}\right) = B_0(e^{w^* - w_0} - 1) + H''(w^*) + \Delta''(w^*),$$

whose modulus is at most  $0.00652e^{0.00652/5} + \sup |H''| + 10^{-60} < 0.0135$ , since  $B_0|e^{w^* - w_0} - 1| \leq B_0|w^* - w_0|e^{|w^* - w_0|}$ . Dividing by  $B_0 - 1/c$  and taking  $\frac{1}{2} \log(1 + \cdot)$  gives  $|T_4| \leq 0.0068/(B_0 - 1/c)$ .

$T_5$  is the remainder of Theorem 6.1, evaluated at the true saddle rather than dropped. Here  $B_0 \geq 5$  forces  $w_0 \geq 5.3492$  and  $w^* \geq w_0 - 0.0014$ , so  $e^{w^*} \geq 210.1$  and  $|T_5| \leq \sum_{k \geq 0} 2e^{-2 \cdot 3^k \cdot 210.1} < 10^{-182}$ .

Each bound is  $O(1/B_0)$  (for  $T_4$ , via  $B_0 - 1/c \geq B_0 - 1$ , comparable to  $B_0$  once  $B_0 \geq 5$ ). Finally  $\tau = w_0 - \log B_0$  and  $w_0 = c(B_0 + a)$  give  $\tau = cB_0 + ca - \log B_0$ , so  $B_0/\tau \rightarrow 1/c$ , and  $O(1/B_0) = O(1/\tau)$  with constants free of the phase of  $\tau$ .  $\square$

## 9 The Berg–Krüppel identity

**Proposition 9.1.** *Write  $f(p) = \alpha \log p - \beta \log^2 p$ , so that Berg and Krüppel's transform  $G_0(p) = p^\alpha \exp(-\beta \log^2 p)$  of their equation (9.3) is  $e^{f(p)}$  and their equation (9.5) reads*

$$g_0(t) = \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} \exp[pt + f(p)] dp, \quad \sigma > 0.$$

*Take  $\alpha, \beta$  from their equation (9.4) at  $a = 3$ ,  $\lambda = 2/3$ . Then  $f(p) = Q(\log p)$  identically; the exact saddle of that integral,  $t + f'(x) = 0$  with  $t = e^{-\tau}$ , is  $x^\sharp = e^{w_0(\tau)}$ ; and*

$$P(\tau) = x^\sharp t + f(x^\sharp) - \frac{1}{2} \log(2\pi f''(x^\sharp)),$$

*the saddlepoint expression invoked in the proof of their Proposition 9.1, evaluated at that exact saddle. In the variable  $Y := cB_0$ , with  $r = -ca - \log c$ , the saddle equation is  $Y - \log Y = \tau + r$ .*

*Proof.* Their truncated equation (9.2) is  $\lambda \psi'(t) = a \psi(at)$  for  $a > 1$ ,  $\lambda > 0$ ; Laplace transforming (the boundary term  $\lambda \psi(0)$  from  $\mathcal{L}[\psi'](p) = pG(p) - \psi(0)$  vanishes, since the solutions of interest are supported on  $[0, \infty)$  with  $\psi(0) = 0$ ) gives  $\lambda p G(p) = G(p/a)$ , whose general solution (9.3) is  $G_0$  times an arbitrary 1-periodic function of  $\log p / \log a$ , with

$$\alpha = -\frac{1}{2} - \frac{\log \lambda}{\log a}, \quad \beta = \frac{1}{2 \log a}$$

their equation (9.4). Section 5 placed  $\varphi$  in that family at  $a = 3$ ,  $\lambda = 2/3$ : their untruncated (9.1) reads  $\frac{2}{3} \varphi'(t) = 3(\varphi(3t) - \varphi(3t - 2))$ , which on  $(0, \frac{2}{3})$  is  $\varphi'(t) = \frac{9}{2} \varphi(3t)$ . At those values

$$\alpha = -\frac{1}{2} - \frac{\log(2/3)}{\log 3} = \frac{1}{2} - \frac{\log 2}{\log 3} = a, \quad \beta = \frac{1}{2 \log 3} = \frac{1}{2c},$$

where the  $a$  on the right is this paper's linear coefficient of  $Q$ ; their  $a$ , the dilation factor 3, is a different quantity and is not written again below. Hence, writing  $y := \log p$ ,

$$f(p) = \alpha y - \beta y^2 = ay - \frac{y^2}{2c} = Q(y),$$

the first of the proposition's four claims. This is an identity in  $y$ , not merely a value match: the chain rule ( $p d/dp = d/dy$ ) applied to  $f$ 's first and second derivatives confirms it also at the level of  $Q'$  and  $Q''$ ,

$$-p f'(p) = -Q'(y) = B_{\text{sm}}(y) = 2\beta y - \alpha, \quad p^2 f''(p) = Q''(y) - Q'(y) = B_{\text{sm}}(y) - \frac{1}{c} = 2\beta y - 2\beta - \alpha.$$

Berg and Krüppel's own displayed formula for  $f''(p)$ , on p. 179 of [3], in the proof of their Proposition 9.1 (the unnumbered display right after "In view of"), reads  $p^2 f''(p) = 2\beta \ln p + 2\beta - \alpha$ , with  $+2\beta$  where direct differentiation of  $f$  gives  $-2\beta$ :  $f''(p) = \frac{1}{p^2}(2\beta \ln p - 2\beta - \alpha)$ , since  $f'(p) = \frac{1}{p}(\alpha - 2\beta \ln p)$  differentiates to  $-\frac{2\beta}{p^2} - \frac{\alpha - 2\beta \ln p}{p^2}$ . Their Proposition 9.1 is unaffected, the term being of lower order there than  $2\beta \log p$ , but the exact expression used here is not: the  $\pm 2\beta$  discrepancy shifts  $(x^\sharp)^2 f''(x^\sharp) = B_0 - 1/c$  by  $2/c$ , hence shifts  $-\frac{1}{2} \log(2\pi(B_0 - 1/c))$  in  $P(\tau)$  below by  $O(1/B_0)$ , which vanishes as  $B_0 \rightarrow \infty$ .

The saddle condition  $t + f'(x) = 0$  at  $x = e^y$  now reads  $e^{-\tau} = B_{\text{sm}}(y)e^{-y}$ , that is  $\tau = y - \log B_{\text{sm}}(y) = \Phi_0(y)$ , whose unique solution on the branch  $B_{\text{sm}} > 1/c$  is  $y = w_0(\tau)$  by Lemma 8.1. So  $x^\sharp = e^{w_0}$ , and there  $x^\sharp t = e^{w_0 - \tau} = B_0$ ,  $f(x^\sharp) = Q(w_0)$ ,  $(x^\sharp)^2 f''(x^\sharp) = B_0 - 1/c > 0$ . Substituting,

$$\begin{aligned} x^\sharp t + f(x^\sharp) - \frac{1}{2} \log(2\pi f''(x^\sharp)) &= B_0 + Q(w_0) - \frac{1}{2} \log \frac{2\pi(B_0 - 1/c)}{(x^\sharp)^2} \\ &= Q(w_0) + B_0 + w_0 - \frac{1}{2} \log\left(2\pi(B_0 - \frac{1}{c})\right), \end{aligned}$$

which is  $P(\tau)$ . Finally  $w_0 = c(B_0 + a)$  and  $\tau = w_0 - \log B_0$  give  $\tau = cB_0 + ca - \log B_0 = Y - \log Y + ca + \log c = Y - \log Y - r$ .  $\square$

Series-reverting  $Y - \log Y = \tau + r$  to second order: writing  $W := \tau + r$ , a first substitution  $Y = W + \log W + \delta$  into  $Y - \log Y = W$  gives, since  $\log(1+u) = u + O(u^2)$  with  $u = (\log W + \delta)/W$ ,  $\delta = (\log W + \delta)/W + O(\log^2 W/W^2)$ , so  $\delta = \log(W)/W + O(\log^2 W/W^2)$  (the single term  $Y = W + \log W$  alone is only accurate to  $O(\log W/W)$ , one order too coarse; this second term is needed). Hence  $Y = W + \log W + \log(W)/W + O(\log^2 W/W^2)$ , and  $\log W = \log \tau + r/\tau + O(1/\tau^2)$  then gives  $Y = \tau + \log \tau + r + (\log \tau + r)/\tau + O(\tau^{-2} \log^2 \tau)$ . Since  $w_0 = Y + ca$  and  $Q(w_0) = -Y^2/(2c) + ca^2/2$  (substituting  $w_0 = Y + ca$  into  $Q$ 's definition and simplifying),  $P$ 's leading term in  $Y$  is  $-Y^2/(2c)$ , so  $\partial P/\partial Y = -Y/c + O(1)$  turns the  $O(\tau^{-2} \log^2 \tau)$  error in  $Y$  above into the  $O(\tau^{-1} \log^2 \tau)$  error in  $P$  below. Substituting into  $P$  gives

$$\begin{aligned} P(\tau) &= -\frac{(\tau + \log \tau)^2}{2c} + \left(1 + \frac{1}{c} - \frac{r}{c}\right)\tau + \left(\frac{1}{2} - \frac{r}{c}\right)\log \tau + C_P + O\left(\frac{\log^2 \tau}{\tau}\right), \\ C_P &= -\frac{r^2}{2c} + r + ca + \frac{ca^2}{2} - \frac{\log 2\pi}{2} + \frac{\log c}{2} = -0.9576743183133982760669122497855855 \dots \end{aligned}$$

Berg and Krüppel's own equations following (9.6) give, in their notation,

$$\delta_{\text{BK}} = \frac{1}{2} + \alpha - 2\beta \log(2\beta), \quad \gamma = -2\beta - \delta_{\text{BK}} - \frac{1}{2}, \quad \varepsilon = \frac{1}{2} + \alpha - \beta \log(2\beta),$$

with  $\alpha = a$ ,  $\beta = 1/(2c)$  as identified above. Substituting and simplifying (using  $r = -ca - \log c$ ) gives, matching the  $\tau$ ,  $\log \tau$ , and constant coefficients of  $P$  term for term,

$$\gamma = -\left(1 + \frac{1}{c} - \frac{r}{c}\right) = -1.8649154949 \dots, \quad \delta_{\text{BK}} = \frac{1}{2} - \frac{r}{c} = 0.4546762683 \dots,$$

$$\varepsilon = a + \frac{1}{2} + \frac{\log c}{2c} = 0.4118732573 \dots,$$

so that, in particular,  $C_P = \varepsilon \log(2\beta) - \frac{1}{2} \log(2\pi)$  holds identically (verified by substituting  $\varepsilon$ 's expression directly into  $C_P$ 's), giving

$$C_P = \varepsilon \log(2\beta) - \frac{1}{2} \log(2\pi) = \log \frac{(2\beta)^\varepsilon}{\sqrt{2\pi}}.$$

Hence  $P(\tau) - \log \varphi_{0,\text{bare}}(e^{-\tau}) \rightarrow C_P$  and, using the prefactor-included normalization,  $P(\tau) - \log \varphi_0(e^{-\tau}) \rightarrow 0$  with rate  $O(\tau^{-1} \log^2 \tau)$ : the  $\tau$ ,  $\log \tau$ , and constant coefficients of  $P$  reproduce  $\gamma$ ,  $\delta_{\text{BK}}$ ,  $\varepsilon$  exactly, so this convergence is a direct consequence of the identity of Proposition 9.1 together with the series reversion above, not an appeal to Berg and Krüppel's own asymptotic analysis of  $\varphi_0$ .

*Proof of Corollary 10.3.* By Proposition 8.2 and 9.1,  $S(\tau) = \log \varphi_0(e^{-\tau}) + H(w_0(\tau)) + O(\tau^{-1} \log^2 \tau)$ , i.e.  $\varphi_{\text{sp}} = \varphi_0 \cdot e^{H(w_0(\tau))}(1 + o(1))$ ; by Theorem 7.5,  $\varphi = \varphi_{\text{sp}}(1 + o(1))$  as well, so  $\varphi = \varphi_0 \cdot e^{H(w_0(\tau))}(1 + o(1))$ : the periodic factor multiplying Berg and Krüppel's asymptotic solution, in equation (9.7) for this eigenfunction, is  $e^H$ , matching the product formula their Proposition 9.3 supplies for it, as noted after Theorem 6.1. Proposition 6.3's Fourier series for  $H$  is the new, closed form; Proposition 6.5's certificate is what neither their product formula nor their closing remark (that they only *expect*  $\varphi$  bounded and bounded away from zero relative to  $\varphi_0$ ) settles.  $\square$

## 10 Wirsching's Conjecture 3, and the failure of the unrestricted asymptotic

Wirsching reaches Conjecture 3 by retreating from a stronger statement. He needs to replace  $\varphi$  by  $\varphi_0$  under the limit in [1, eq. (7.5)], and writes:

“That replacement would be possible, if, e.g.,  $\varphi \sim c \varphi_0$  for some constant  $c > 0$ . It turns out that the following (slightly weaker) conjecture about the asymptotics of  $\varphi$  suffices to imply (★4):”

The conjecture that follows is (★5) of §3, restated here for reference, thirty pages after it was first met. His constant  $c$  is kept as he wrote it and is unrelated to  $c := \log 3$  of §6.

*Wirsching's Conjecture 3.* There are positive real constants  $c$  and  $\delta_5$  such that

$$\lim_{\ell \rightarrow \infty} \frac{\varphi(z_\ell)}{\varphi_0(z_\ell)} = c > 0 \quad \text{uniformly for sequences } (z_\ell) \in \tilde{A}_{\delta_5}.$$

Both statements are settled below. The weaker one holds and the stronger one fails.

**Theorem 10.1** (Wirsching's Conjecture 3). *Wirsching's Conjecture 3 is true. For each  $\delta_5 > 0$ , on his comparison class  $\tilde{A}_{\delta_5}$ , the ratio  $\varphi(z_\ell)/\varphi_0(z_\ell)$  converges uniformly to  $e^{H(0)}$ , a well-defined real number by Theorem 6.1 regardless of its decimal expansion, evaluated here as  $e^{H(0)} = 0.534122 \dots$  (a floating-point evaluation; Proposition 6.5's proof certifies the nearby difference  $H(0) - H(\log \frac{3}{2})$  rigorously, but not  $H(0)$  itself in isolation), where  $H$  is the periodic function of Theorem 6.1 and the normalization of  $\varphi_0$  is Berg and Krüppel's own, equation (9.6) of [3], written out at (2). Conjecture 3 itself only asserts the existence of some positive constant, independently of normalization; the specific value  $e^{H(0)}$  is tied to this particular normalization of  $\varphi_0$ . The rate depends on  $\delta_5$ .*

**Theorem 10.2** (Failure of the unrestricted asymptotic). *The stronger asymptotic  $\varphi(t) \sim \kappa \varphi_0(t)$  as  $t \rightarrow 0^+$ , for any single constant  $\kappa$  and with no restriction to a comparison class, is false. The ratio  $\varphi(t)/\varphi_0(t)$  oscillates indefinitely as  $t \rightarrow 0^+$ , with amplitude bounded below by a positive constant: it is governed by the exactly  $c$ -periodic function  $H$  evaluated at the saddle location  $w_0(\tau)$ ,  $\tau = \log(1/t)$ , whose phase modulo  $c$  sweeps every value infinitely often as  $\tau \rightarrow \infty$  (Lemma 8.1). That the amplitude is positive follows from a classical theorem, with no computation; its explicit numeric size is a computer-assisted result, a rigorous interval-arithmetic certificate (Proposition 6.5).*

One computation underlies both. Write  $t = e^{-\tau}$  and let  $H(w)$  be the  $\log 3$ -periodic function of Theorem 6.1; the saddlepoint bridge of §§7–9 gives  $\log \varphi(t) - \log \varphi_0(t) = H(w_0(\tau)) + o(1)$ , where the saddle location advances by  $c$  per period of  $\tau$ , up to a vanishing correction:  $w_0(\tau + c) - w_0(\tau) = c + O(1/\tau)$  (Lemma 8.1). Wirsching’s comparison class constrains  $w_0(\tau_\ell)$ ’s phase modulo  $\log 3$  to converge to a single value as  $\ell \rightarrow \infty$ ; the unrestricted range  $\tau \rightarrow \infty$  with no class restriction instead sweeps that phase through every value, since  $\Phi_0$  is a bijection (again Lemma 8.1). The two theorems read one object at two scopes.

**Corollary 10.3.**  *$e^H$  is Berg and Krüppel’s own periodic factor for the eigenfunction  $\varphi$ , from Proposition 9.3 of [3], which they give as an infinite product;  $H$  itself is its logarithm. To our knowledge its Fourier series, Proposition 6.3, has not previously been given for this function<sup>2</sup>. This corollary has two parts of different depth: that  $e^H$  agrees with their product as a function, proved in Remark 6.2 by pure algebra ( $e^H = e^{-Q}e^L e^{-\Delta}$  recombining Theorem 6.1’s own pieces), and that this function is the periodic factor multiplying  $\varphi$ ’s own asymptotic, which needs the full saddlepoint chain of Theorem 7.5 and Propositions 8.2–9.1.*

Throughout the rest of this section, Theorem 7.5’s parameter  $N$  and Sections 8–9’s parameter  $\tau$  are combined freely into one  $o(1)$ : since  $s = e^{w^*}$  and, by Lemma 8.1,  $w^* = w_0(\tau) + O(1/B_0)$  with  $w_0(\tau) = c(B_0 + a)$ ,  $N = \lfloor \log_3(2s) \rfloor = \tau/c + O(\log \tau)$  as  $\tau \rightarrow \infty$ , so  $N \rightarrow \infty$  exactly when  $\tau \rightarrow \infty$ . On  $A_{\delta_5}$  below,  $\tau_\ell \rightarrow \infty$  uniformly, at a rate depending only on  $\ell$  and  $\delta_5$ , not on the particular sequence chosen.

**Lemma 10.4** (Phase locking on the comparison class). *Fix  $\delta_5 > 0$  and let  $(z_\ell) \in \tilde{A}_{\delta_5}$ , with  $\tau_\ell := -\log z_\ell$  and  $\lambda_\ell := 3^\ell z_\ell / \ell$ . Then  $\lambda_\ell \rightarrow 1$  and*

$$w_0(\tau_\ell) - \ell c = -\log \lambda_\ell + O(1/\ell),$$

*uniformly on the class. Consequently  $\text{dist}(w_0(\tau_\ell), c\mathbb{Z}) = O(\ell^{-1/2})$  uniformly, and  $H(w_0(\tau_\ell)) \rightarrow H(0)$ .*

*Proof.* Wirsching’s class  $\tilde{A}_{\delta_5}$  (his equations (1.5) and (3.2)) consists of sequences  $(x_\ell)$  with  $\lfloor 3^\ell x_\ell \rfloor = k_\ell$  satisfying  $|\ell - k_\ell| \leq \delta_5 \sqrt{\ell}$  for every  $\ell$ ; since  $|3^\ell x_\ell - k_\ell| < 1$  as well,  $|3^\ell x_\ell - \ell| < 1 + \delta_5 \sqrt{\ell}$ , hence  $\lambda_\ell \rightarrow 1$  uniformly at rate  $O(1/\ell) + O(\delta_5/\sqrt{\ell})$  on the class. His own  $z_\ell$  from the conjecture statement and his  $x_\ell$  from the class definition coincide, by definition of  $\lambda_\ell$ , so  $\tau_\ell = \ell c - \log \ell - \log \lambda_\ell$ .

For the saddle location, Section 9’s reversion of  $Y - \log Y = \tau + r$  gives  $w_0(\tau) = \tau + \log(\tau/c) + (\log \tau + r)/\tau + O(\tau^{-2} \log^2 \tau)$ . The naive one-term truncation  $w_0(\tau) \approx \tau + \log(\tau/c)$  alone is only accurate to  $O(\log \tau / \tau)$ , which already gives everything Theorem 10.1 uses, since  $\log \ell / \ell = o(\ell^{-1/2})$ ;

<sup>2</sup>Fourier coefficients of a log-periodic fluctuation, obtained as residues of a Dirichlet series or by Mellin inversion, are standard for radix-rational and  $q$ -regular sequences; see [35] and [36], and [28] and [29] for the partition-function case the technique started in. Berg and Krüppel returned to this family after 1998 [33, 34]; those papers treat eigenfunctions, Appell polynomials, addition theorems and generating functions, and contain no Fourier expansion, Mellin transform or  $\zeta$  at all. To our knowledge no closed Fourier expansion has previously been given for the periodic factor of an eigenfunction of this family.

the second term sharpens the lemma to  $O(1/\ell)$  rather than rescuing it. Substituting  $\tau = \tau_\ell$  into the full two-term formula, the  $O(\log \ell/\ell)$  pieces from expanding  $\log(\tau_\ell/c)$  against  $\tau_\ell = \ell c - \log \ell - \log \lambda_\ell$  cancel exactly against the reversion's own  $(\log \tau_\ell + r)/\tau_\ell$  term, leaving the stated  $O(1/\ell)$ , the same cancellation mechanism as the series reversion itself.

Write  $w_0(\tau_\ell) = \ell c + \varepsilon_\ell$  with  $\varepsilon_\ell = -\log \lambda_\ell + O(1/\ell)$ . Then  $\varepsilon_\ell = O(\ell^{-1/2})$ , since  $\log \lambda_\ell = O(\ell^{-1/2})$  by the first paragraph, and once  $|\varepsilon_\ell| < c/2$  the nearest multiple of  $c$  to  $w_0(\tau_\ell)$  is  $\ell c$  itself, so  $\text{dist}(w_0(\tau_\ell), c\mathbb{Z}) = |\varepsilon_\ell| = O(\ell^{-1/2})$ . Every bound here depends on  $\delta_5$  alone. Since  $H$  is  $c$ -periodic and Lipschitz by (10),  $H(w_0(\tau_\ell)) = H(\varepsilon_\ell) \rightarrow H(0)$ , regardless of the sign of  $\varepsilon_\ell$ .  $\square$

*Proof of Theorem 10.1.* By Theorem 7.5, Proposition 8.2 and Proposition 9.1 (the last giving  $P(\tau) - \log \varphi_0(e^{-\tau}) \rightarrow 0$ , Section 9),

$$\log \varphi(z_\ell) - \log \varphi_0(z_\ell) = H(w_0(\tau_\ell)) + o(1)$$

uniformly on  $\tilde{A}_{\delta_5}$ , and Lemma 10.4 sends the right side to  $H(0)$ , uniformly. So  $\varphi(z_\ell)/\varphi_0(z_\ell) \rightarrow e^{H(0)}$  uniformly on the class. This is exactly Conjecture 3, with the constant given explicitly.  $\square$

*Remark 10.5.* The proof above uses only  $\lambda_\ell \rightarrow 1$ , equivalently  $|\ell - k_\ell| = o(\ell)$ ; Wirsching's specific  $O(\sqrt{\ell})$  bound in  $\tilde{A}_{\delta_5}$  plays no role beyond forcing this weaker condition. The same conclusion, and the same limit  $e^{H(0)}$ , holds for the broader class of sequences satisfying only  $|\ell - k_\ell| = o(\ell)$ .

*Proof of Theorem 10.2.* As  $t = e^{-\tau}$  ranges over  $(0, 1/2)$  with no restriction,  $\tau \rightarrow \infty$ . By Lemma 8.1,  $w_0 = \Phi_0^{-1}$  is continuous and strictly increasing and unbounded on  $(\tau_0, \infty)$ , so  $w_0(\tau) \bmod c$  sweeps every value in  $[0, c)$  infinitely often as  $\tau \rightarrow \infty$  (a continuous, strictly increasing, unbounded function hits every residue class modulo  $c$  infinitely often), so by the same combination of results used above,  $\log \varphi(t) - \log \varphi_0(t) = H(w_0(\tau)) + o(1)$  takes values arbitrarily close to both  $\sup H$  and  $\inf H$  infinitely often. By Proposition 6.5,  $\text{osc}(H) > 0$  rigorously, so  $\varphi(t)/\varphi_0(t)$  does not converge, and, writing  $\Psi(t) := \varphi(t)/\varphi_0(t)$ ,

$$\frac{\limsup_{t \rightarrow 0^+} \Psi(t)}{\liminf_{t \rightarrow 0^+} \Psi(t)} \geq e^{\text{osc}(H)} \geq 1.0004188 :$$

the ratio's limsup exceeds its liminf by a factor of at least 1.0004188, i.e. by at least 0.0418%.  $\square$

**Proposition 10.6** (Wirsching's actual requirement). *For  $(x_\ell) \in \tilde{A}_{\delta_5}$ , write  $x_\ell^+ := x_\ell + 3^{-(\ell+1)}$  (his own definition, unnumbered prose on p. 16 of [1], in the quotient estimate that yields his (7.5); since  $x_\ell \sim \ell \cdot 3^{-\ell}$ ,  $x_\ell^+$  is asymptotic to  $x_\ell$  and plays the same role as this paper's  $z_\ell$ ). Wirsching's own use of Conjecture 3, via  $\varphi'(x) = \frac{9}{2}\varphi(3x)$  on  $(0, \frac{2}{3})$  and his own calculation, his equation (7.13), that  $\lim_\ell (2/3^{\ell+1})\varphi'_0(x_\ell)/\varphi_0(x_\ell) = 2/3$  uniformly on the class, reduces to*

$$\limsup_\ell \Lambda_\ell < \frac{3}{2}, \quad \Lambda_\ell := \frac{\varphi(3x_\ell^+)/\varphi_0(3x_\ell^+)}{\varphi(x_\ell^+)/\varphi_0(x_\ell^+)}.$$

*This holds with a wide margin:  $\Lambda_\ell \rightarrow 1$ .*

*Proof.* First, the reduction itself. Wirsching's actual requirement is his inequality (7.5),

$$\limsup_\ell \frac{2}{3^{\ell+1}} \cdot \frac{\varphi'(x_\ell^+)}{\varphi(x_\ell^+)} < 1;$$

by  $\varphi' = \frac{9}{2}\varphi(3\cdot)$  this is  $\limsup_\ell (9/3^{\ell+1})\varphi(3x_\ell^+)/\varphi(x_\ell^+) < 1$ . By definition of  $\Lambda_\ell$ ,  $\varphi(3x_\ell^+)/\varphi(x_\ell^+) = \Lambda_\ell \cdot \varphi_0(3x_\ell^+)/\varphi_0(x_\ell^+)$ , and Berg and Krüppel's  $\varphi_0$  is an asymptotic solution of the same truncated



equation (their equation (7.12) in Wirsching's notation:  $\varphi'_0(t)/\varphi_0(3t) \rightarrow 9/2$  as  $t \rightarrow 0$ , proved from (2) in Remark 11.2), so  $\varphi_0(3x_\ell^+)/\varphi_0(x_\ell^+) = \frac{2}{9}\varphi'_0(x_\ell^+)/\varphi_0(x_\ell^+)(1 + o(1))$ . To see that (7.13) transfers from  $x_\ell$  to  $x_\ell^+$  with an  $o(1)$  correction, write  $v = \log t$ ; the explicit form of  $\varphi_0$  above gives  $\varphi'_0(t)/\varphi_0(t) = B(v)/t$  with  $B(v) = \gamma + \delta_{\text{BK}}/v - 2\beta(v - \log(-v))(1 - 1/v)$ , and, as  $v \rightarrow -\infty$ ,  $B(v) \sim -2\beta v$  while  $B'(v) \rightarrow -2\beta$ , so  $(\log B)'(v) = B'(v)/B(v) \rightarrow 0$  and  $(\log[\varphi'_0/\varphi_0])'(v) = (\log B)'(v) - 1 \rightarrow -1$ , bounded. Since  $x_\ell^+/x_\ell = 1 + 3^{-(\ell+1)}/x_\ell \rightarrow 1$  (as  $x_\ell \sim \ell \cdot 3^{-\ell}$ , in fact  $3^{-(\ell+1)}/x_\ell = O(1/\ell)$ ),  $v(x_\ell^+) - v(x_\ell) = \log(x_\ell^+/x_\ell) = O(1/\ell)$ , and the mean value theorem gives  $\log[\varphi'_0/\varphi_0(x_\ell^+)] - \log[\varphi'_0/\varphi_0(x_\ell)] = O(1/\ell)$ , i.e.  $\varphi'_0(x_\ell^+)/\varphi_0(x_\ell^+) = \varphi'_0(x_\ell)/\varphi_0(x_\ell) \cdot (1 + O(1/\ell))$ . Substituting and using his equation (7.13),  $\lim_\ell (2/3^{\ell+1})\varphi'_0(x_\ell)/\varphi_0(x_\ell) = 2/3$  uniformly on the class,

$$\frac{9}{3^{\ell+1}} \cdot \frac{\varphi(3x_\ell^+)}{\varphi(x_\ell^+)} = \Lambda_\ell \cdot \frac{2}{3^{\ell+1}} \cdot \frac{\varphi'_0(x_\ell^+)}{\varphi_0(x_\ell^+)} (1 + o(1)) = \frac{2}{3} \Lambda_\ell \cdot (1 + o(1)),$$

so the requirement  $\limsup_\ell (9/3^{\ell+1})\varphi(3x_\ell^+)/\varphi(x_\ell^+) < 1$  is exactly  $\limsup_\ell \Lambda_\ell < 3/2$ .

For the value of  $\Lambda_\ell$ : write  $\tau_\ell := -\log x_\ell^+$  (locally to this proof only, distinct from the  $\tau_\ell$  of Theorem 10.1's proof), so  $t = x_\ell^+$  corresponds to  $\tau = \tau_\ell$  and  $t = 3x_\ell^+$  to  $\tau = \tau_\ell - c$ . The phase shift under  $t \mapsto 3t$ , i.e.  $\tau \mapsto \tau - c$ , is  $w_0(\tau - c) - w_0(\tau) + c = O(1/\tau)$  by Lemma 8.1's defining equation, independently of the phase of  $\tau$ ; since  $H$  is  $c$ -periodic,  $H(w_0(\tau - c)) = H(w_0(\tau - c) + c) = H(w_0(\tau) + O(1/\tau))$ , and  $H$  Lipschitz (by (10)) turns this into  $H(w_0(\tau - c)) = H(w_0(\tau)) + O(1/\tau)$ . By Theorem 7.5, Proposition 8.2, and Proposition 9.1 together (the same chain used in the proof of Theorem 10.1),  $\log \varphi(t) - \log \varphi_0(t) = H(w_0(\tau)) + o(1)$  uniformly in phase, so

$$\log \Lambda_\ell = H(w_0(\tau_\ell - c)) - H(w_0(\tau_\ell)) + o(1) = O(1/\tau_\ell) + o(1) \rightarrow 0,$$

using that  $H$  is Lipschitz ((10)) and the phase shift above is  $O(1/\tau_\ell)$ ; hence  $\Lambda_\ell \rightarrow 1$ , well inside  $\limsup \Lambda_\ell < 3/2$ .  $\square$

## 11 The chain, closed to a single condition

Theorem 10.1 settles ( $\star 5$ ). Wirsching carries ( $\star 5$ ) to ( $\star 4$ ) himself, in [1, §7]. Composing the two puts ( $\star 4$ ) on the same footing as his own two theorems, and Theorem 4.3 has already removed the conjectural link below ( $\star 3$ ). One condition is left.

**Theorem 11.1** (Condition ( $\star 4$ )). *Condition ( $\star 4$ ) holds, for every window radius  $\delta > 0$ , with  $\mu = 1/3$ .*

*Proof.* Proposition 10.6 already gives this. Wirsching's requirement is his inequality (7.5), and that proposition shows (7.5) is exactly  $\limsup_\ell \Lambda_\ell < 3/2$  and that  $\Lambda_\ell \rightarrow 1$ , on  $\tilde{A}_{\delta_5}$  for every  $\delta_5 > 0$ . So (7.5) holds with a wide margin, and it holds at every radius rather than at one produced by a construction.

Wirsching's own route reaches the same place through his statement of Conjecture 3, and is worth writing out, both because it is the argument a reader of [1] will have in hand and because it fixes the constant. Write  $x_\ell^+ := x_\ell + 3^{-\ell-1}$  [1, §7]. Bounding the transition kernel above and below by  $\varphi$ , the source reduces ( $\star 4$ ) to

$$\limsup_{\ell \rightarrow \infty} \frac{2}{3^{\ell+1}} \cdot \frac{\varphi'(x_\ell^+)}{\varphi(x_\ell^+)} \leq 1 - \mu < 1, \quad (x_\ell) \in \tilde{A}_\delta, \quad (18)$$

its equation (7.5). Fix  $\delta > 0$  and choose  $\delta_5 > \delta$ . By [1, eq. (7.14)] there are sequences  $(\hat{x}_\ell), (\hat{y}_\ell) \in \tilde{A}_{\delta_5}$  with  $\hat{x}_\ell = x_\ell^+$  and  $\hat{y}_{\ell-1} = 3x_\ell^+$  for all but finitely many  $\ell$ . Then

$$\frac{2}{3^{\ell+1}} \cdot \frac{\varphi'(x_\ell^+)}{\varphi(x_\ell^+)} = \frac{2}{3^{\ell+1}} \cdot \frac{9}{2} \frac{\varphi(\hat{y}_{\ell-1})}{\varphi(\hat{x}_\ell)} = \frac{2}{3^{\ell+1}} \cdot \frac{9}{2} \frac{\varphi_0(\hat{y}_{\ell-1})}{\varphi_0(\hat{x}_\ell)} \cdot \frac{\varphi(\hat{y}_{\ell-1})/\varphi_0(\hat{y}_{\ell-1})}{\varphi(\hat{x}_\ell)/\varphi_0(\hat{x}_\ell)},$$

the first equality by  $\varphi'(x) = \frac{9}{2}\varphi(3x)$  on  $[0, 2/3]$  [1, eq. (7.7)]. Theorem 10.1 sends both ratios in the last factor to  $e^{H(0)} > 0$ , uniformly on  $\tilde{A}_{\delta_5}$ , so that factor tends to 1: the numerator is controlled by the uniform error at index  $\ell - 1$  and the denominator by the one at index  $\ell$ , and the limit is positive. This is the only place Conjecture 3 is used, and the constant cancels, which is why a class-dependent limit suffices.

What remains is  $\frac{2}{3^{\ell+1}} \cdot \frac{9}{2} \varphi_0(3x_\ell^+)/\varphi_0(x_\ell^+)$ . Here  $\varphi_0$  solves the truncated equation  $\lambda g'(t) = a g(at)$  only asymptotically, in the source's own sense  $\lim_{t \rightarrow 0} \lambda \varphi_0'(t)/(a \varphi_0(at)) = 1$  [1, eq. (7.12)], which at  $\lambda = 2/3, a = 3$  reads  $\frac{2}{9} \varphi_0'(t)/\varphi_0(3t) \rightarrow 1$  and which Remark 11.2 proves from (2). Every sequence in  $\tilde{A}_{\delta_5}$  satisfies  $\ell - \delta_5 \sqrt{\ell} \leq \lfloor 3^\ell x_\ell \rfloor \leq 3^\ell x_\ell \leq \lfloor 3^\ell x_\ell \rfloor + 1 \leq \ell + \delta_5 \sqrt{\ell} + 1$ , so  $x_\ell^+$  lies between  $(\ell - \delta_5 \sqrt{\ell})3^{-\ell}$  and  $(\ell + \delta_5 \sqrt{\ell} + 2)3^{-\ell}$ , both bounds depending on  $\delta_5$  alone, so the convergence is uniform on the class and

$$\frac{2}{3^{\ell+1}} \cdot \frac{9}{2} \frac{\varphi_0(3x_\ell^+)}{\varphi_0(x_\ell^+)} = (1 + o(1)) \frac{2}{3^{\ell+1}} \cdot \frac{\varphi_0'(x_\ell^+)}{\varphi_0(x_\ell^+)} \rightarrow \frac{2}{3}$$

by [1, eq. (7.13)], which holds for every  $\delta > 0$  and follows from (2) by differentiation. So the lim sup in (18) is  $2/3$ , giving  $1 - \mu = 2/3$  and  $\mu = 1/3$ .

This second route delivers  $(\star 4)$  only for  $\delta < \delta_5$ , since [1, eq. (7.14)] needs  $\delta_5 > \delta$ . That is already enough on its own terms, because  $(\star 4)$  is existential in its window radius: Conjecture 2 reads “if there are real numbers  $\delta, \mu > 0$  such that...” [1, §6], and one admissible radius is what it asks for. The first route does not need the observation at all.  $\square$

*Remark 11.2* (Steps the source states rather than carries out). The reduction imported above rests on six steps that [1, §7] states without carrying out, one of them asserted without argument by Berg and Krüppel as well. All six are true, and this remark proves them, so that nothing in Proposition 10.6 or Theorem 11.1 rests on an unverified source claim. Throughout,  $K_\ell(x, 0) = \frac{3}{2}(W_3^\ell \chi_0)(x)$  and  $K_\ell(x, 1) = \frac{3}{2}(W_3^\ell \chi_1)(x)$  are the iterated kernels of [1, §7], and  $\varphi$  is the density of  $X = \sum_{j \geq 1} 2U_j 3^{-j}$  as in Proposition 3.1.

*A probabilistic form of  $W_3$ .* Substituting  $t = 3x - 2u$  in  $W_3 f(x) = \frac{3}{2} \int_{3x-2}^{3x} f$  gives  $W_3 f(x) = 3 \int_0^1 f(3x - 2u) du$ , so iterating,

$$(W_3^\ell f)(x) = 3^\ell \mathbb{E} f(3^\ell x - 3^\ell X_\ell), \quad X_\ell := 2 \sum_{j=1}^{\ell} 3^{-j} U_j,$$

the truncation of  $X$  at depth  $\ell$ . Write  $F_\ell$  for its distribution function and, for  $\ell \geq 1$ ,  $\varphi_\ell$  for its density;  $X_0 = 0$  is an atom and  $F_0 = \mathbf{1}_{[0, \infty)}$ . Write  $F_\varphi$  for the distribution function of  $X$ . Since  $\chi_0 = \mathbf{1}_{[0, 2/3]}$  and  $\chi_1 = \mathbf{1}_{[1/3, 1]}$ , for  $\ell \geq 1$  and with  $h := 3^{-\ell}$ ,

$$(W_3^\ell \chi_0)(x) = 3^\ell \int_{x-\frac{2}{3}h}^x \varphi_\ell, \quad (W_3^\ell \chi_1)(x) = 3^\ell \int_{x-h}^{x-\frac{1}{3}h} \varphi_\ell. \quad (19)$$

Two facts about  $X$  are used below. Replacing each  $U_j$  by  $1 - U_j$  sends  $X$  to  $1 - X$ , so  $X$  is symmetric about  $\frac{1}{2}$ . And  $X = \frac{2}{3}U_1 + \frac{1}{3}X'$  with  $X'$  an independent copy gives  $\varphi(v) = \frac{3}{2}(F_\varphi(3v) - F_\varphi(3v - 2))$ ,

whence  $\varphi \leq \frac{3}{2}$ , with  $\varphi < \frac{3}{2}$  on  $(0, \frac{1}{3})$  and so  $F_\varphi(s) < \frac{3}{2}s$  for every  $s > 0$ . The same decomposition of  $X_\ell = \frac{2}{3}U_1 + \frac{1}{3}X'_{\ell-1}$  gives

$$\varphi_\ell(v) = \frac{3}{2}F_{\ell-1}(3v) \quad \text{for } v \leq \frac{1}{3} \quad (\ell \geq 1), \quad (20)$$

since the second term drops there:  $\varphi_\ell$  is nondecreasing on  $(-\infty, \frac{1}{3}]$ .

(7.3),  $K_\ell(x, 0) = K_\ell(x + 3^{-(\ell+1)}, 1)$ . For  $\ell \geq 1$ , immediate from (19): replacing  $x$  by  $x + \frac{1}{3}h$  in the second integral turns its limits into those of the first. At  $\ell = 0$  it is  $\chi_0(x) = \chi_1(x + \frac{1}{3})$ , which is the definition of  $\chi_1$ .

(7.4),  $K_\ell(x, 0) > \varphi(x) > K_\ell(x, 1)$  for  $0 < x < \frac{1}{3}$ . Take  $\ell \geq 1$ ; at  $\ell = 0$  the three quantities are  $\frac{3}{2}$ ,  $\varphi(x)$  and 0. Put  $\psi(v) := \varphi_\ell(x - hv)$  for  $v \in [0, 1]$ , so that (19) reads  $K_\ell(x, 0) = \frac{3}{2} \int_0^{2/3} \psi$  and  $K_\ell(x, 1) = \frac{3}{2} \int_{1/3}^1 \psi$ , while conditioning  $X = X_\ell + hX'$  on an independent copy  $X' \in [0, 1]$  gives  $\varphi(x) = \int_0^1 \psi(v)\varphi(v)dv$ . All three are therefore averages of  $\varphi_\ell$  over the same window  $[x - h, x]$ , which lies in  $(-\infty, \frac{1}{3}]$  because  $x < \frac{1}{3}$ ; by (20)  $\psi$  is nonincreasing there. Writing a nonincreasing  $\psi \geq 0$  as a mixture of indicators  $\mathbf{1}_{[0,s]}$ , both inequalities reduce to statements about  $F_\varphi$  at every  $s \in (0, 1]$ : the first to  $F_\varphi(s) \leq \frac{3}{2} \min(s, \frac{2}{3})$ , the second to  $F_\varphi(s) \geq \frac{3}{2} \max(0, s - \frac{1}{3})$ . The first is  $F_\varphi(s) \leq \frac{3}{2}s$ , which is  $\varphi \leq \frac{3}{2}$ ; the second is the first evaluated at  $1 - s$ , by the symmetry of  $\varphi$  about  $\frac{1}{2}$ . Both gaps are in fact strict at every  $s \in (0, 1)$ :  $F_\varphi(s) < \frac{3}{2}s$  there, and  $1 - F_\varphi(s) > 0$ , which by the same reflection covers the second. It remains to see that the layer measure charges such an  $s$ , that is, that  $\psi(0) > \psi(1)$ . By (20) that reads  $F_{\ell-1}(3x) > F_{\ell-1}(3x - 3^{-(\ell-1)})$ . For  $\ell \geq 2$  it holds because  $X_{\ell-1}$  has a density positive on the interior of  $[0, 1 - 3^{-(\ell-1)}]$ , and  $(3x - 3^{-(\ell-1)}, 3x]$  meets that interior in a set of positive length whenever  $0 < 3x < 1$ . For  $\ell = 1$  the atom gives  $\psi(0) = \frac{3}{2}$  and  $\psi(1) = 0$  directly.

The unlabelled Taylor step  $\varphi(x - h) \geq \varphi(x + h) - 2h\varphi'(x + h)$  needs  $\varphi$  convex near 0, which holds because differentiating  $\varphi'(x) = \frac{9}{2}\varphi(3x)$  gives  $\varphi''(x) = \frac{27}{2}\varphi'(3x) \geq 0$  while  $3x$  stays in the region where  $\varphi$  is nondecreasing.

(7.12), the asymptotic solution property  $\lambda\varphi'_0(t)/(a\varphi_0(at)) \rightarrow 1$ . Both Wirsching and Berg and Krüppel state this without argument, and it carries Proposition 10.6 and both routes to Theorem 11.1. It follows from (2). Write  $L := -\log t \rightarrow \infty$ , so that (2) becomes  $\log \varphi_0(t) = C - \gamma L + \delta_{\text{BK}} \log L - \beta(L + \log L)^2$  with  $C = \varepsilon \log(2\beta) - \frac{1}{2} \log(2\pi)$ , and

$$\log \varphi'_0(t) = \log \varphi_0(t) + L + \log\left(\gamma - \frac{\delta_{\text{BK}}}{L} + 2\beta(L + \log L)\left(1 + \frac{1}{L}\right)\right).$$

Subtract the first display at  $3t$ , where  $L$  becomes  $L - c$ . The linear term contributes  $-\gamma c$ ; the logarithmic one  $\delta_{\text{BK}} \log \frac{L}{L-c} = O(1/L)$ ; the bracket of the second display contributes  $\log(2\beta) + \log L + O(\log L/L)$ . For the quadratic one put  $A = L + \log L$  and  $B = L - c + \log(L - c)$ : then  $A - B = c + c/L + O(L^{-2})$  and  $A + B = 2L + 2\log L - c + O(1/L)$ , so  $A^2 - B^2 = 2cL + 2c\log L - c^2 + 2c + O(\log L/L)$  and, since  $2\beta c = 1$  and  $\beta c^2 = c/2$ , that term contributes  $-L - \log L + \frac{c}{2} - 1$ . The  $L$  and  $\log L$  pieces cancel against  $L$  and  $\log L$  above, leaving

$$\log \frac{\varphi'_0(t)}{\varphi_0(3t)} \longrightarrow -\log c - \gamma c + \frac{c}{2} - 1 = \log \frac{9}{2}, \quad (21)$$

the last equality by substituting  $\gamma = -2\beta - \delta_{\text{BK}} - \frac{1}{2}$  and  $\beta = 1/(2c)$ ; the two sides agree to every one of 120 computed digits. At  $\lambda = 2/3$ ,  $a = 3$  this is exactly (7.12). The limit depends on  $t$  only through  $L$ , so it is uniform over any family of  $t$  with  $-\log t \rightarrow \infty$  uniformly, which is what  $\tilde{A}_{\delta_5}$  supplies. As a check, evaluating the left side of (21) numerically from (2) at  $t = \ell 3^{-\ell}$  gives 1.50373, 1.5040772 and 1.504077393 at  $\ell = 10^2, 10^4, 10^5$ , against  $\log \frac{9}{2} = 1.504077397 \dots$

(7.13), introduced with “a somewhat lengthy calculation shows”, follows from (2) by differentiation. With  $v = \log t$ ,  $\varphi'_0(t)/\varphi_0(t) = B(v)/t$  where  $B(v) = \gamma + \delta_{\text{BK}}/v - 2\beta(v - \log(-v))(1 - \frac{1}{v}) = -2\beta v(1 + O(\log|v|/|v|))$ . For  $(x_\ell) \in \tilde{A}_\delta$ ,  $x_\ell = \ell 3^{-\ell}(1 + O(\delta/\sqrt{\ell}))$  and  $v = -\ell c + O(\log \ell)$ , so  $\frac{2}{3^{\ell+1}}\varphi'_0(x_\ell)/\varphi_0(x_\ell) = \frac{2}{3} \cdot \frac{2\beta\ell c}{\ell}(1 + o(1)) \rightarrow \frac{2}{3}$  using  $2\beta c = 1$ . Every error term depends on  $\delta$  alone, so the limit is uniform on the class, which is what (7.13) asserts.

(7.14), *the class lifting*. Given  $(x_\ell) \in \tilde{A}_\delta$  and  $\delta_5 > \delta$ , the sequences  $\hat{x}_\ell := x_\ell^+$  and  $\hat{y}_{\ell-1} := 3x_\ell^+$  lie in  $\tilde{A}_{\delta_5}$  for all but finitely many  $\ell$ . Indeed  $\lfloor 3^\ell x_\ell^+ \rfloor = \lfloor 3^\ell x_\ell + \frac{1}{3} \rfloor \in \{k_\ell, k_\ell + 1\}$  where  $k_\ell = \lfloor 3^\ell x_\ell \rfloor$ , and  $\lfloor 3^{\ell-1}(3x_\ell^+) \rfloor$  is that same integer, so the window conditions at index  $\ell$  and at index  $\ell - 1$  both follow from  $|k_\ell - \ell| \leq \delta\sqrt{\ell}$  once  $\delta\sqrt{\ell} + 2 \leq \delta_5\sqrt{\ell - 1}$ , which holds for all large  $\ell$  because  $\delta_5 > \delta$ .

*One attribution*. Wirsching gives the penultimate equality of the calculation above to his (7.14) and (7.7). His (7.7) is an identity for  $\varphi$ , and the line it is applied to carries  $\varphi_0$ ; what licenses that step is (7.12), which contributes a factor  $1 + o(1)$  rather than an equality. None of the six points needs a hypothesis the source does not already have, and none changes  $\mu = 1/3$ .

Corollary 4.4 now reads as a statement about the whole chain. Condition  $(\star 3)$  implies uniform positive predecessor density on the non-cyclic integers  $a \not\equiv 0 \pmod 3$  through proved implications only, with no conjecture among them.

State what that does and does not say. Conditions  $(\star 3)$ ,  $(\star 2)$  and  $(\star 1)$  remain unproved: Theorem 4.3 proves the implication  $(\star 2) \Rightarrow (\star 1)$  and not its hypothesis, and Wirsching’s Theorem 2 proves  $(\star 3) \Rightarrow (\star 2)$  and not its hypothesis. What has changed is that no conjecture stands between them: a proof of  $(\star 3)$  now propagates through two proved implications to  $(\star 2)$  and  $(\star 1)$ , and so to the conclusion. That is one direction only. Should  $(\star 3)$  turn out false,  $(\star 2)$  could still hold, and so could  $(\star 1)$ . What follows is that  $(\star 3)$  is the only condition whose truth still has to be established in order to complete the chain.

Wirsching’s route to  $(\star 3)$  is Conjecture 2, the implication  $(\star 4) \Rightarrow (\star 3)$ , whose antecedent Theorem 11.1 now supplies. It is not the only conceivable route: a proof of  $(\star 3)$  by any other means closes the chain just as well, which is why “only Conjecture 2 remains” is the wrong summary. §12 says what is known about the implication itself.

*Remark 11.3* (What Theorem 10.2 does not reach). The failure of the unrestricted asymptotic costs the chain nothing. None of  $(\star 1)$  through  $(\star 4)$  mentions  $\varphi_0$ : the first three are stated in terms of the generators, their Haar averages and the Elka functions, and the fourth in terms of  $W_3$  acting on  $\chi_0$  and  $\chi_1$ . Within [1, §7],  $\varphi_0$  occurs in the statement of  $(\star 5)$ , in (7.11) where it is defined, in (7.12), in (7.13), in the proof of Theorem 11.1 above, and in the sentence quoted at the head of §10, where he raises  $\varphi \sim c\varphi_0$  unrestricted and declines it. That last one is exactly what Theorem 10.2 refutes, which is the point: he did not use it. Of the rest, (7.12) is an unrestricted limit as  $t \rightarrow 0$  but says nothing about the ratio  $\varphi/\varphi_0$ , and the others are restricted to a comparison class or to a fixed sequence tending to 0 at the borderline rate. Nothing the chain uses needs the unrestricted form.

*Remark 11.4* (Why the retreat was enough). Theorem 10.2 explains why Wirsching could not keep  $\varphi \sim c\varphi_0$ . It does not explain why the weaker conjecture sufficed, and the calculation above does. The two points at which the ratio cancels are  $\hat{x}_\ell = x_\ell^+$  and  $\hat{y}_{\ell-1} = 3x_\ell^+$ , a factor of 3 apart. Multiplying the argument by 3 shifts  $\tau = \log(1/t)$  by  $c$ , and  $H$  has period  $c$ , so the two periodic corrections agree and cancel whether or not  $H$  is constant. The agreement is asymptotic, since  $H$  is read at the saddle location and Lemma 8.1 gives only  $w_0(\tau + c) - w_0(\tau) = c + O(1/\tau)$ ; that is enough, because the cancellation is already inside a limit. The rigorous form of this is the second half of Proposition 10.6’s proof, where  $\log \Lambda_\ell = H(w_0(\tau_\ell - c)) - H(w_0(\tau_\ell)) + o(1)$  and the first difference is  $O(1/\tau_\ell)$  by periodicity and (10). Anything built on that pair of points inherits the

same weakness. The quantity in his (7.5) is a ratio of  $\varphi$  at  $3x_\ell^+$  and at  $x_\ell^+$ , and it carries the periodic correction only through  $H(w_0(\tau_\ell - c)) - H(w_0(\tau_\ell))$ , which (10) bounds by a constant times the residual phase shift. At  $x_\ell = \ell 3^{-\ell}$  with  $\ell = 500$ , so that  $\tau_\ell = -\log x_\ell^+ = 543.09086\dots$ , that shift is  $w_0(\tau_\ell - c) - w_0(\tau_\ell) + c = -2.0051362\dots \times 10^{-3}$ , and (10) bounds the difference by  $\sup_w |H'(w)|$  times its modulus, which is  $< 2.402 \times 10^{-6}$ .

## 12 Discussion

Two of Wirsching's five conditions are now established outright,  $(\star 5)$  by Theorem 10.1 and  $(\star 4)$  by Theorem 11.1, and every implication from  $(\star 3)$  downward is a theorem, so  $(\star 3)$  implies uniform positive predecessor density (Corollary 4.4). Conditions  $(\star 3)$ ,  $(\star 2)$  and  $(\star 1)$  are all still unproved.  $(\star 3)$  carries the other two through proved implications, in that direction only, so it is the only one whose truth still has to be established to complete the chain. Conjecture 2 is the source's route into it and is not the only conceivable one.

Wirsching stated the weaker Conjecture 3 instead of the stronger  $\varphi \sim \kappa\varphi_0$  he considers first. Theorem 10.2 shows why: the stronger statement fails by a mechanism with a certified, positive amplitude. Every phase  $\theta \in [0, c)$  swept by  $H$  admits some sequence along which the ratio converges to  $e^{H(\theta)}$ : take  $\tau_\ell := \Phi_0(\theta + \ell c)$ ,  $z_\ell := e^{-\tau_\ell}$ , so  $w_0(\tau_\ell) = \theta + \ell c$  exactly, and apply Theorem 7.5, Proposition 8.2, and Proposition 9.1 as in the proof of Theorem 10.1. What is special about phase 0 is only that it is the one Wirsching's own class  $\tilde{A}_{\delta_5}$  happens to fix.

Remark 11.4 answers the other half of the question, why the weaker statement was enough. Wirsching's proof does one thing with the constant, which is to cancel it across a factor of 3, and a factor of 3 is one period of  $H$ . His argument never sees the periodic correction at all.

What remains is  $(\star 3)$ , and establishing the antecedent leaves its difficulty exactly where it was. The asymmetry recorded in §3 has two parts. Condition  $(\star 4)$  carries no residue-coordinate information of its own, so any proof of  $(\star 4) \Rightarrow (\star 3)$  has to recover that information from further structure in Wirsching's generators rather than from the hypothesis. Neither bridge the source offers from  $(\star 4)$  toward  $(\star 3)$  supplies that information uniformly in growing resolution. Wirsching's own analytic route of two years earlier, described in §2, ran into the same analytic-to-combinatorial gap.

An explicit rate in Theorem 10.1 is not pursued here. The sharper Edgeworth correction of Remark 7.7, and a correspondingly sharper form of the envelope and Berg–Krüppel comparisons, would give a polynomial rate throughout; neither theorem needs it.

## Data Availability Statement

Every computation this paper reports is backed by a script in one repository, <https://github.com/faculdade/wirsching-conjectures-1-and-3>, archived at doi:10.5281/zenodo.22116346, release v1.0.1, commit c0339d1. That is the version DOI, which is fixed to this exact snapshot; the concept DOI doi:10.5281/zenodo.22116225 always resolves to the latest version and is not what a reader reproducing this paper wants. The repository is organized as one folder per section of this paper and numbered as this paper numbers them, with a README per folder stating what its scripts verify, how to run them, and what to expect. Sections 1, 2, 3 and 5 carry no computation. The archive also carries a copy of this paper, as it stood at the release.

Two kinds of claim here are computer-assisted, and both are in the folder for §6: the enclosure of  $\text{osc}(H)$  in Proposition 6.5 and the derivative bounds (10). These use Arb ball arithmetic through python-flint at a working precision stated at the top of the script, higher than the 250-bit precision

quoted inline in Proposition 6.5’s proof, so they are an independent higher-precision reproduction of that certificate rather than a lower-precision approximation of it. Each prints its input balls and its final enclosure, so the printed run output is the certificate. The non-constancy of  $H$  is not among them: it is proved analytically from the zero-free theorem for  $\zeta$  on  $\operatorname{Re} s = 1$ , with no computation.

Everything else is exact integer or rational arithmetic, or high-precision floating arithmetic through mpmath. Where a script claims an inequality it asserts it and exits non-zero on failure. The repository also carries the numerical work that preceded Theorem 10.1, including an evaluation of  $\varphi$  through depth  $\ell = 500$  with zero truncation error at every point tested. No claim in this paper rests on it, and its folder says so.

Two earlier repositories preceded this one and remain where they are, with their DOIs valid: <https://github.com/faculdade/wirsching-conjecture3-proof>, archived at doi:10.5281/zenodo.21854549, commit f8248c3, for the standalone Conjecture 3 preprint; and <https://github.com/faculdade/collatz-wirsching-2003>, for the unsubmitted manuscript that carried Conjecture 1. Their scripts are carried into the repository above, renumbered to this paper’s sections. The first numbers its results as they were first published; the new repository’s README carries the mapping between the two numberings.

## Relation to earlier versions of this work

Wirsching’s Conjecture 1, §4 here, first circulated in an unsubmitted manuscript, together with material on Conjecture 2 not carried over. Conjecture 3, §§5 to 10 here, appeared as a preprint under the title *Wirsching’s Conjecture 3, and the periodic correction to Berg–Krüppel’s density asymptotic*. The present paper supersedes both. §11 is new here and appears in neither.

## Declaration of Generative AI and AI-assisted technologies

During the preparation of this work the author used Claude (Anthropic) and Codex (OpenAI) as research assistants: for literature search, for developing and auditing the accompanying code, for adversarial review of the arguments, and for drafting and revising prose. All source-dependent claims were checked by the author against primary sources. The author independently reviewed all mathematical arguments and computational outputs and takes full responsibility for the content.

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